

Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

1. Introduction
 2. Spin-orbital angular momentum interaction in light
 3. Light-matter interaction
 4. Exciton wave packet
- Summary and outlook

1. Introduction

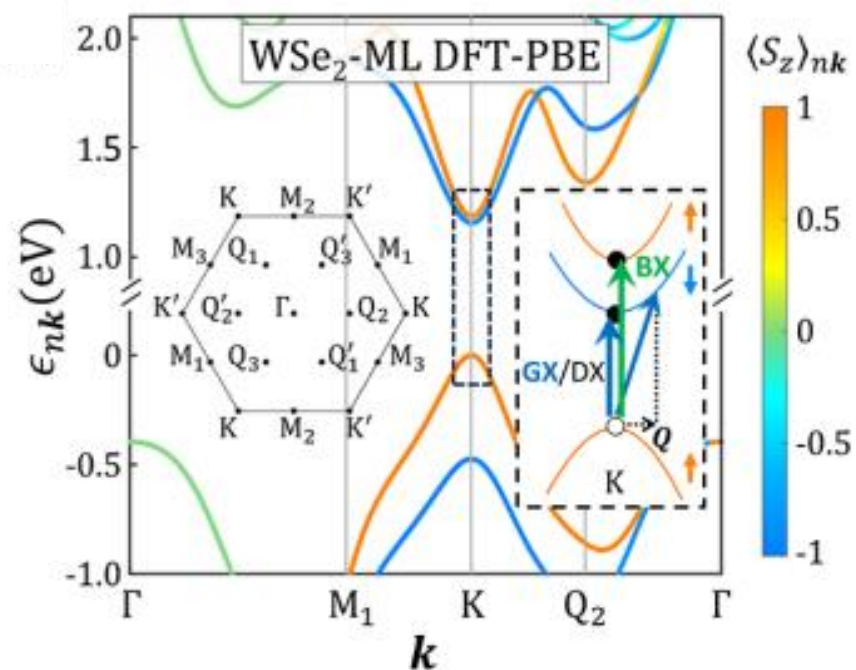
2. Spin-orbital angular momentum interaction in light

3. Light-matter interaction

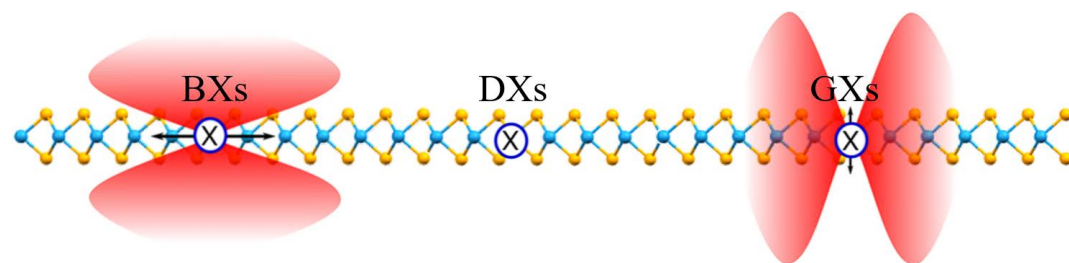
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Summary and outlook

1. Introduction

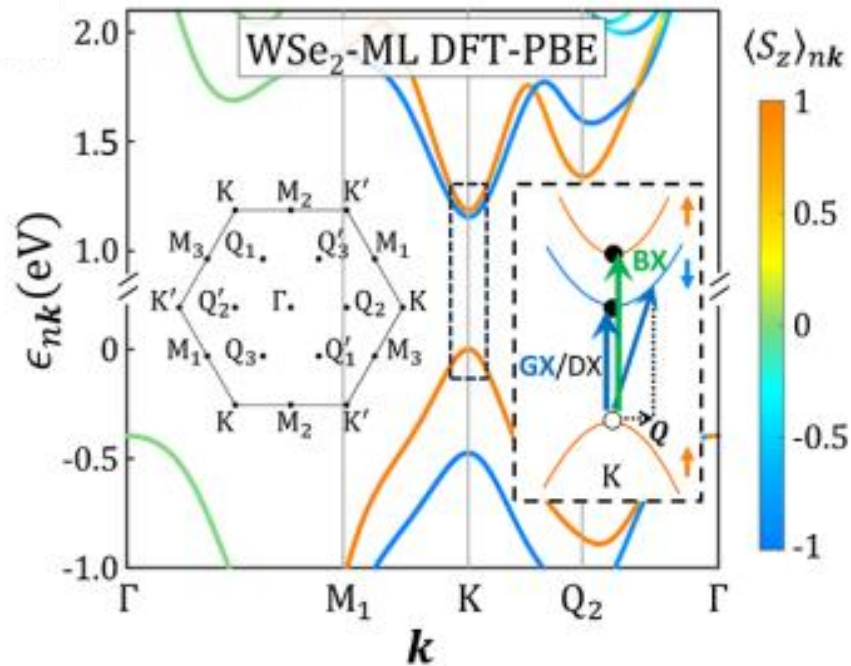


- Spin-resolved calculated WSe₂ band structure
O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024)

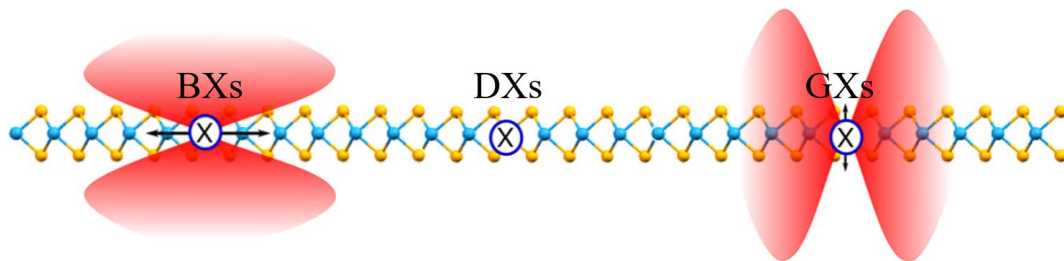


- Excitons categorized by radiation patterns
L. M. Schneider *et al.*, *Sci. Rep.* **10**, 8091 (2020)

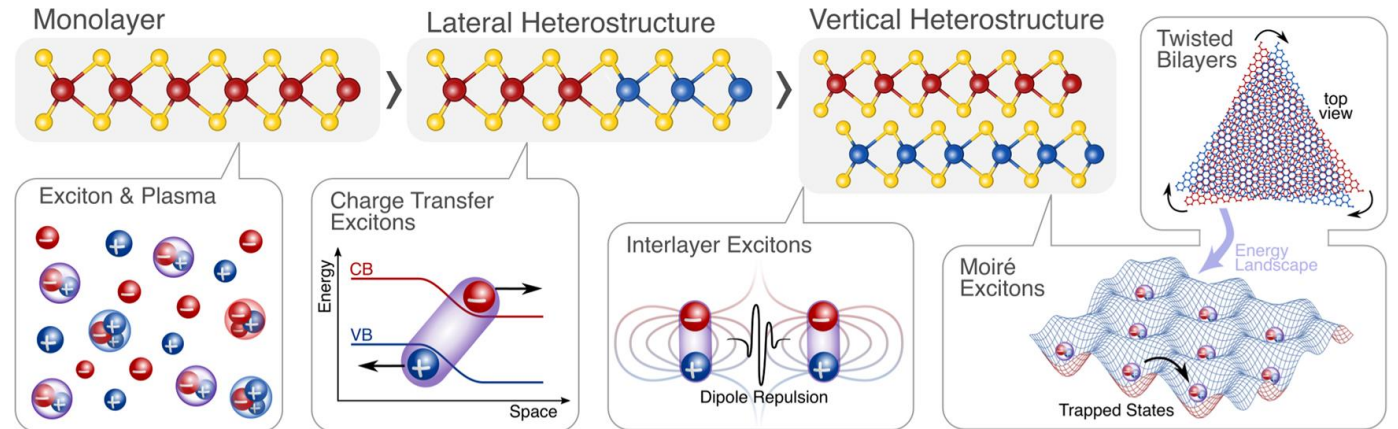
1. Introduction



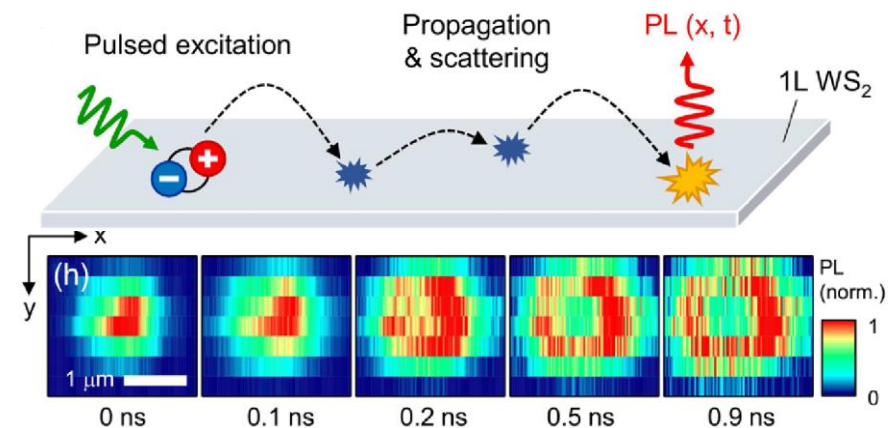
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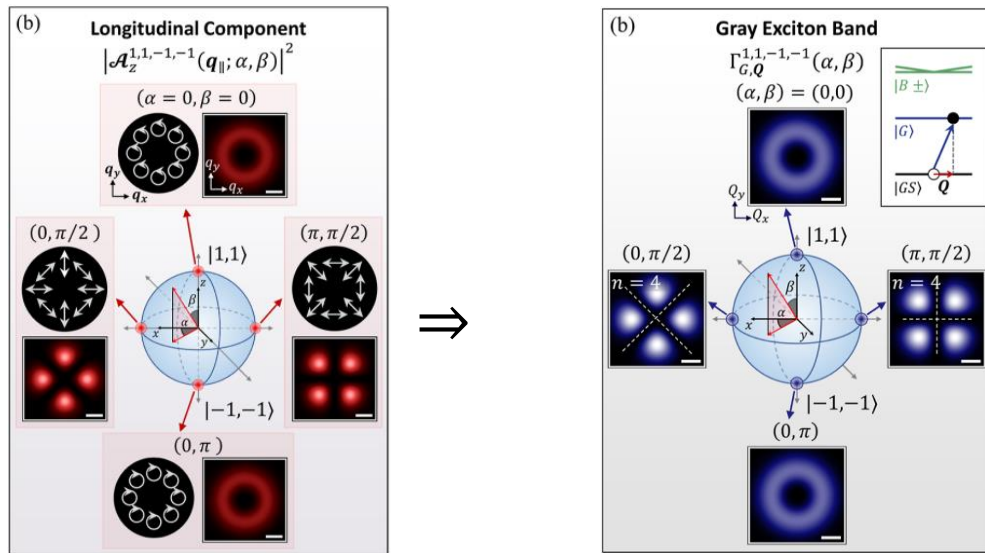


- Exciton transport behavior in different transition metal dichalcogenide (TMD) material systems
E. Malic *et al.*, *Nat. Commun.* **14**, 3430 (2023)

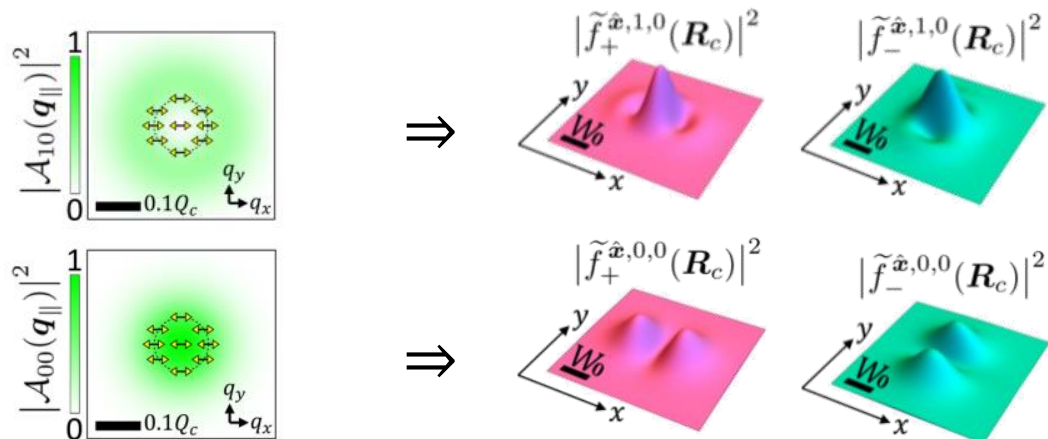


- Exciton propagate in SiO₂-supported WS₂
M. Kulig *et al.*, *PRL* **120**, 207401 (2018)

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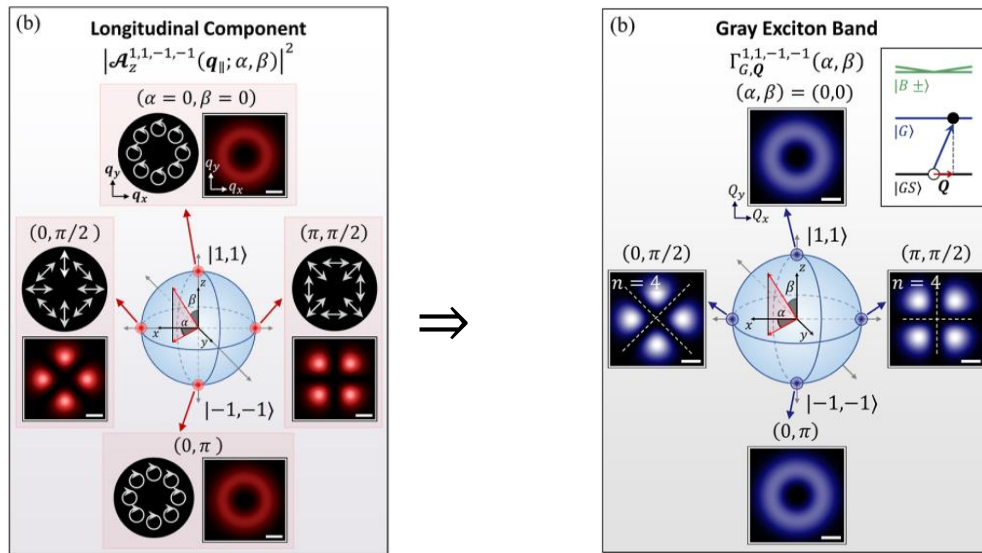


- Transition rate of gray exciton excited by vector vortex beam
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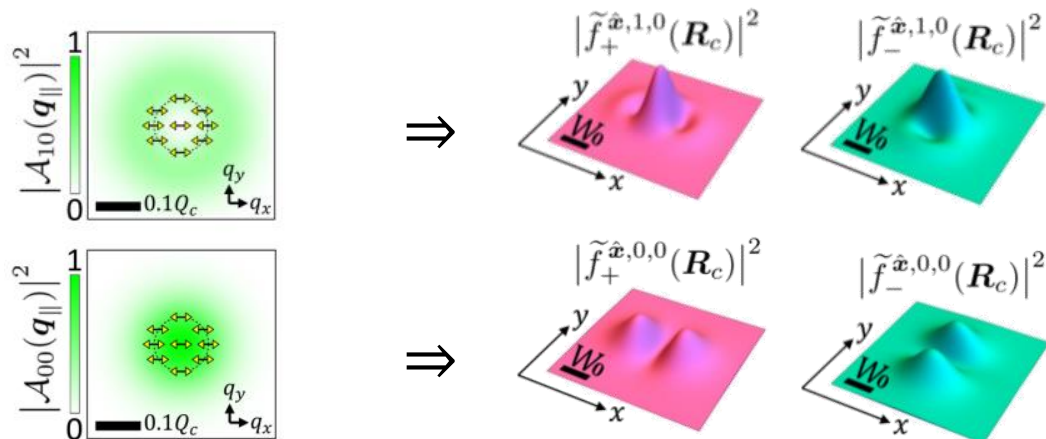
- Wave packet of valley-mixed bright exciton excited by Laguerre-Gaussian beam
G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019)

1. Introduction



Further cases study

- Transition rate of gray exciton excited by vector vortex beam
 O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024)



Global phase problem study

- Wave packet of valley-mixed bright exciton excited by Laguerre-Gaussian beam
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1. Introduction

2. Spin-orbital angular momentum interaction (SOI) in light

- Laguerre-Gaussian beams
- Longitudinal component : SOI effect
- Longitudinal component : Vector vortex beams

3. Light-matter interaction

4. Exciton wave packet

Summary and outlook

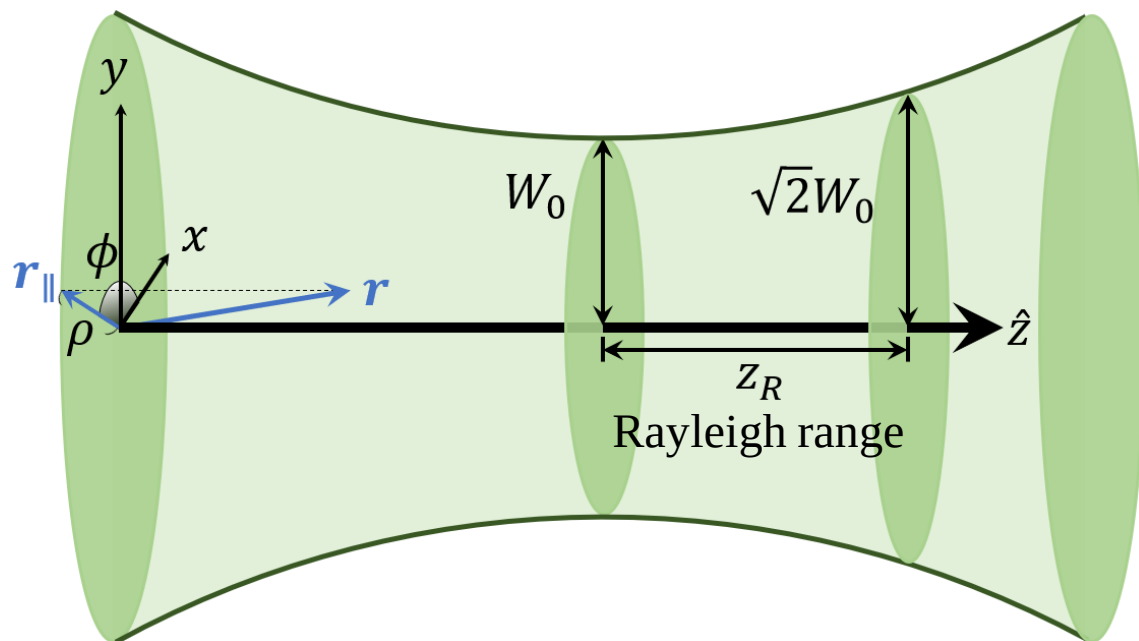
2. Laguerre-Gaussian (LG) beam : Distribution

LG beam is solved as vector potential $\mathbf{A}(\mathbf{r})$ from paraxial Helmholtz equation within Lorentz gauge.

With **long Rayleigh range approximation** :

$$A_{\sigma,\ell,p}^L(\mathbf{r}) = A_{\ell,p}^L(\mathbf{r}) \hat{\mathbf{e}}_{\sigma} \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z} \hat{\mathbf{e}}_{\sigma}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$



- Polarization unit vector :

$$\hat{\mathbf{e}}_{\sigma} = \hat{\mathbf{x}} + i\sigma\hat{\mathbf{y}}, \quad \sigma = \pm 1$$

(LG beam carries SAM of $\sigma\hbar$)

- Topological charge:

$$\ell \in \mathbb{Z}$$

(LG beam carries OAM of $\ell\hbar$)

- Beam waist : W_0
- Rayleigh range : $z_R = \frac{1}{2} q_0 W_0^2$
- Radial mode index: integer $p \geq 0$
- Constant amplitude : A_0^{LG}
- Generalized Laguerre polynomial :

$$L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m \frac{(|\ell|+p)!}{(p-m)! (|\ell|+m)! m!} x^m$$

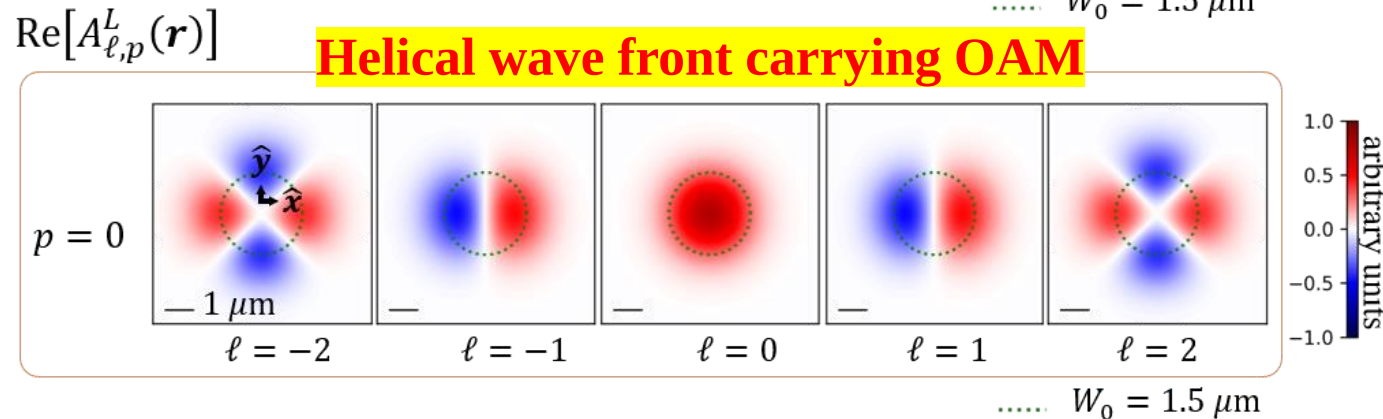
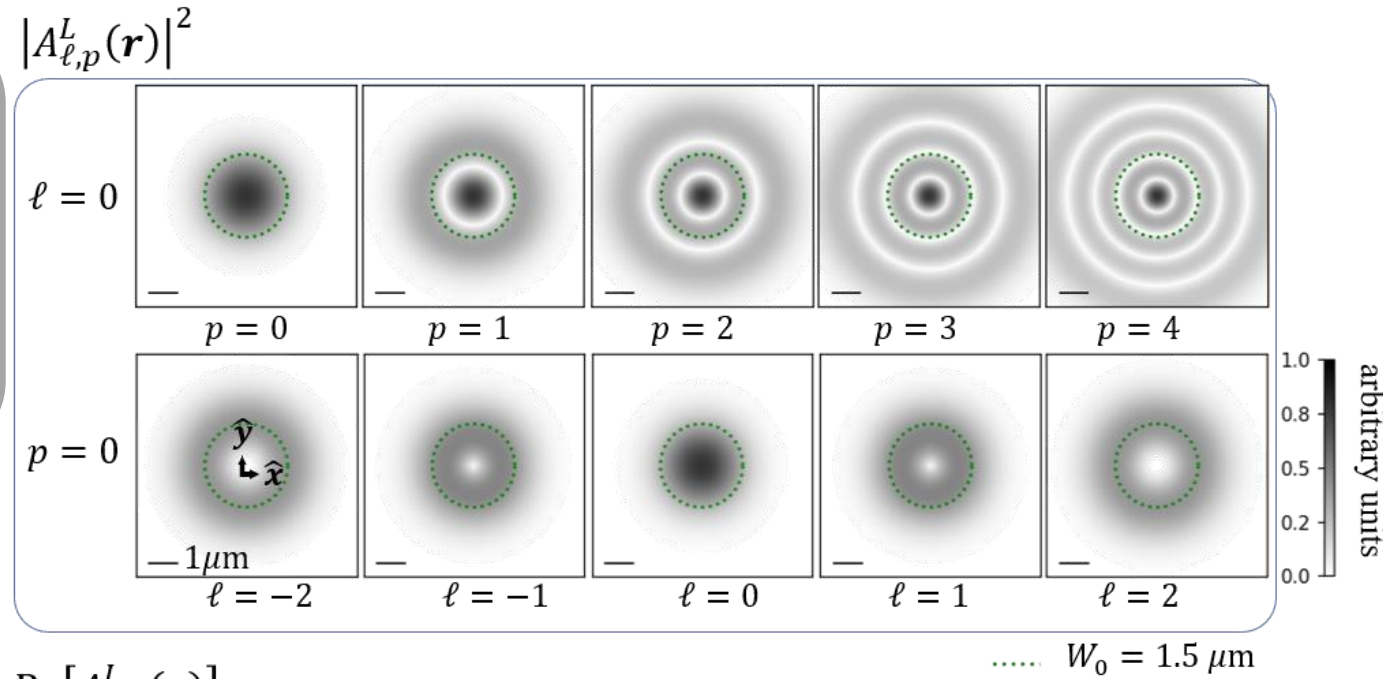
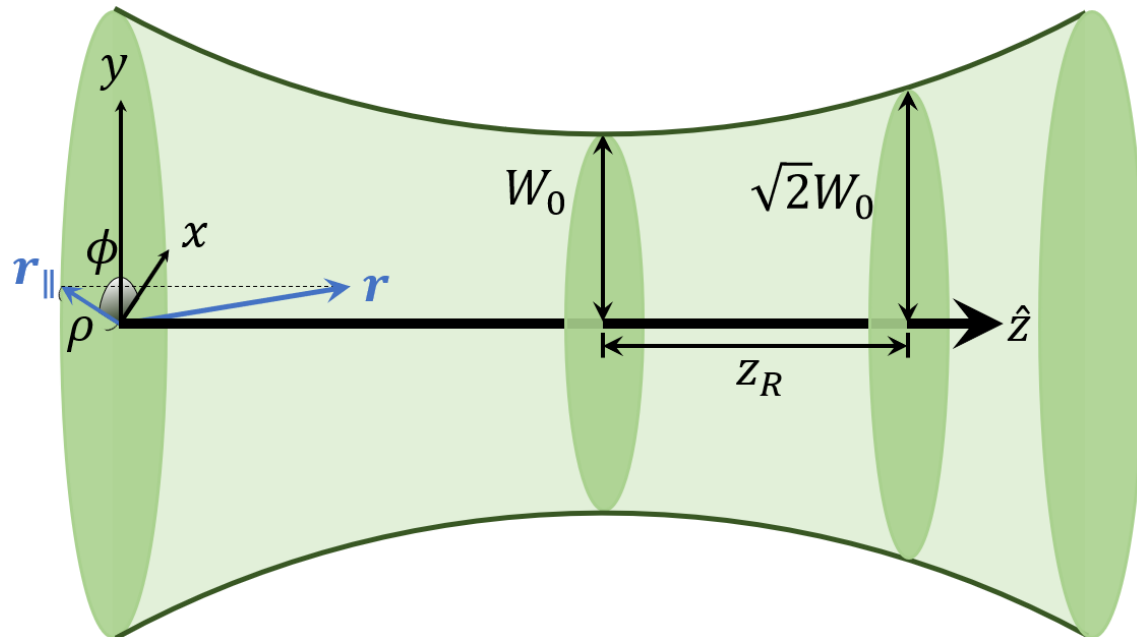
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2. Longitudinal component : SOI effect

LG beam solutions (spectrum / gauge) ($p = 0$):

Real / Lorentz :

$$\mathbf{A}_{\sigma,\ell}^L(\mathbf{r}) \approx f_\ell(\rho) e^{i\ell\phi} e^{iq_0 z} \hat{\mathbf{e}}_\sigma$$

Angular / Lorentz :

$$\mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_\parallel) \approx \tilde{F}_{|\ell|}(q_\parallel) e^{i\ell\phi q_\parallel} \hat{\mathbf{e}}_\sigma$$

Real / Coulomb :

$$\mathbf{A}_{\sigma,\ell,p}(\mathbf{r}) \approx \tilde{A}_\ell^\parallel(\mathbf{r}) \hat{\mathbf{e}}_\sigma + \tilde{A}_{\sigma,\ell}^z(\mathbf{r}) \hat{\mathbf{z}}$$

Angular / Coulomb :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_\parallel) \approx \tilde{\mathcal{A}}_\ell^\parallel(\mathbf{q}_\parallel) \hat{\mathbf{e}}_\sigma + \tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_\parallel) \hat{\mathbf{z}}$$

Change spectrum

Expanded in plane-wave basis

$$\mathcal{A}_{\sigma,\ell}^L(r) = \sum_{\mathbf{q}_\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_\parallel) e^{i\mathbf{q}_\parallel \cdot \mathbf{r}_\parallel} e^{iq_0 \cdot z}$$

Change gauge

Relation from Maxwell's equations

$$\mathbf{A}_{\sigma,\ell}(r) = \mathbf{A}_{\sigma,\ell}^L(r) + \frac{\nabla \left(\nabla \cdot \mathbf{A}_{\sigma,\ell}^L(r) \right)}{q_0^2}$$

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$$\mathbf{A}_{\sigma,\ell}(r) = \mathbf{A}_{\sigma,\ell}^L(r) + \frac{\nabla \left(\nabla \cdot \mathbf{A}_{\sigma,\ell}^L(r) \right)}{q_0^2}$$

Additional SOI component

$$\tilde{A}_{\sigma,\ell}^z(\mathbf{r}) \approx \frac{i}{\sqrt{2}q_0\rho} \left[(|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right] f_\ell(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_\parallel) \approx -\frac{\sin \theta_q}{\sqrt{2}} \tilde{F}_{|\ell|}(q_\parallel) e^{i(\sigma+\ell)\phi_{q_\parallel}}$$

2. Longitudinal component : SOI effect

LG beam solutions Angular / Coulomb ($p = 0$):

Real / Lorentz :

$$\mathbf{A}_{\sigma,\ell}^L(\mathbf{r}) \approx f_\ell(\rho) e^{i\ell\phi} e^{iq_0 z} \hat{\mathbf{e}}_\sigma$$

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Angular / Coulomb :

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What we really need

Change spectrum

Expanded in plane-wave basis

$$\mathcal{A}_{\sigma,\ell}^L(\mathbf{r}) = \sum_{\mathbf{q}_\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_\parallel) e^{i\mathbf{q}_\parallel \cdot \mathbf{r}_\parallel} e^{iq_0 \cdot z}$$

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Relation from Maxwell's equations

$$\mathbf{A}_{\sigma,\ell}(\mathbf{r}) = \mathbf{A}_{\sigma,\ell}^L(\mathbf{r}) + \frac{\nabla \left(\nabla \cdot \mathbf{A}_{\sigma,\ell}^L(\mathbf{r}) \right)}{q_0^2}$$

Theory of light matter interaction

$$\Gamma^{S,\mathbf{Q}} = \frac{2\pi}{\hbar} |\tilde{M}^{S,\mathbf{Q}}|^2 \rho^L(\hbar\omega = E^{S,\mathbf{Q}})$$

$$\tilde{M}^{S,\mathbf{Q}} \approx \frac{E_g}{2i\hbar} \delta_{q_\parallel,\mathbf{Q}} \mathcal{A}(\mathbf{q}_\parallel) \cdot \mathbf{D}^{S,\mathbf{Q}*}$$

2. Longitudinal component : SOI effect

LG beam in **angular spectrum** / **Coulomb gauge** ($p = 0$):

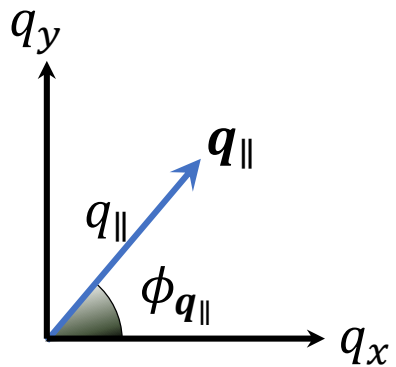
$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel})\hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel})\hat{\mathbf{z}}$$

Transverse component :

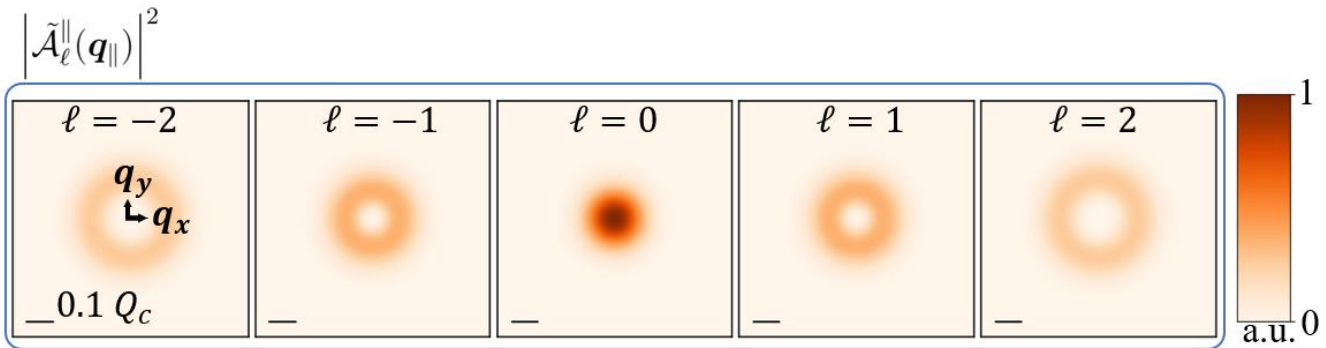
$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel})e^{i\ell\phi_{q_{\parallel}}}$$

Longitudinal component :

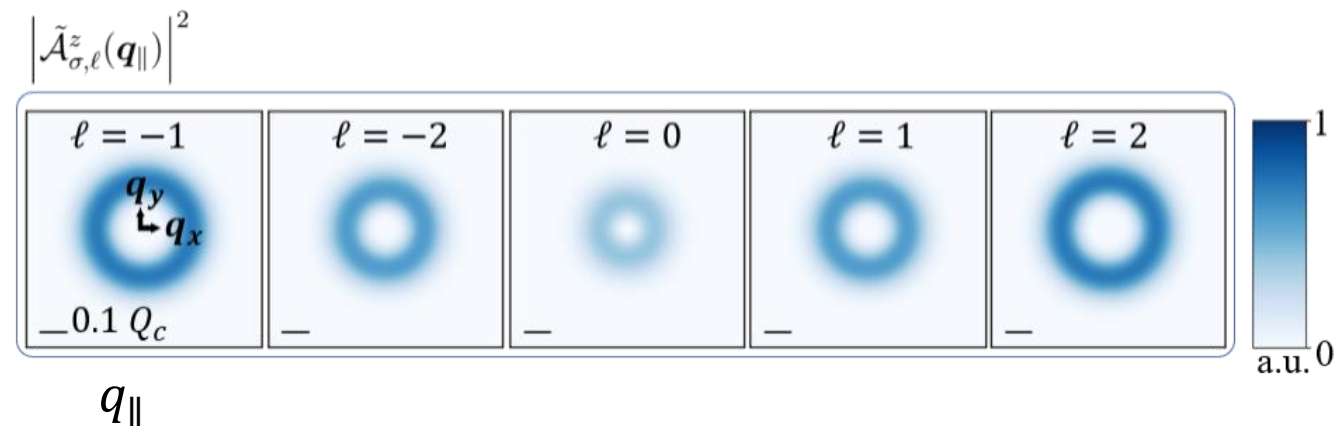
$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin\theta_q}{\sqrt{2}}\tilde{F}_{|\ell|}(q_{\parallel})e^{iJ\phi_{q_{\parallel}}} \quad \text{SOI} \quad J = \sigma + \ell$$



Transverse component :



Longitudinal component :



2. Longitudinal component : SOI effect

LG beam in **angular spectrum** / **Coulomb gauge** ($p = 0$):

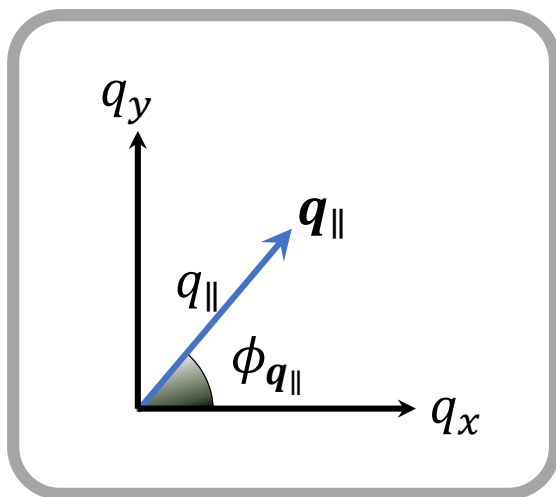
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Transverse component :

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel})e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

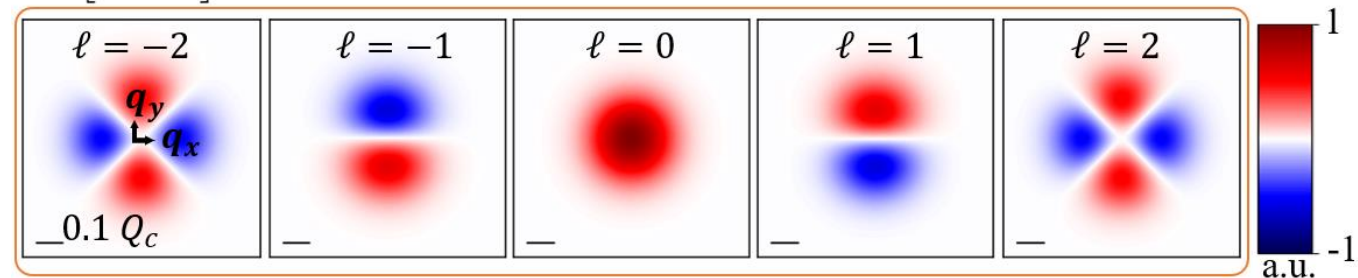
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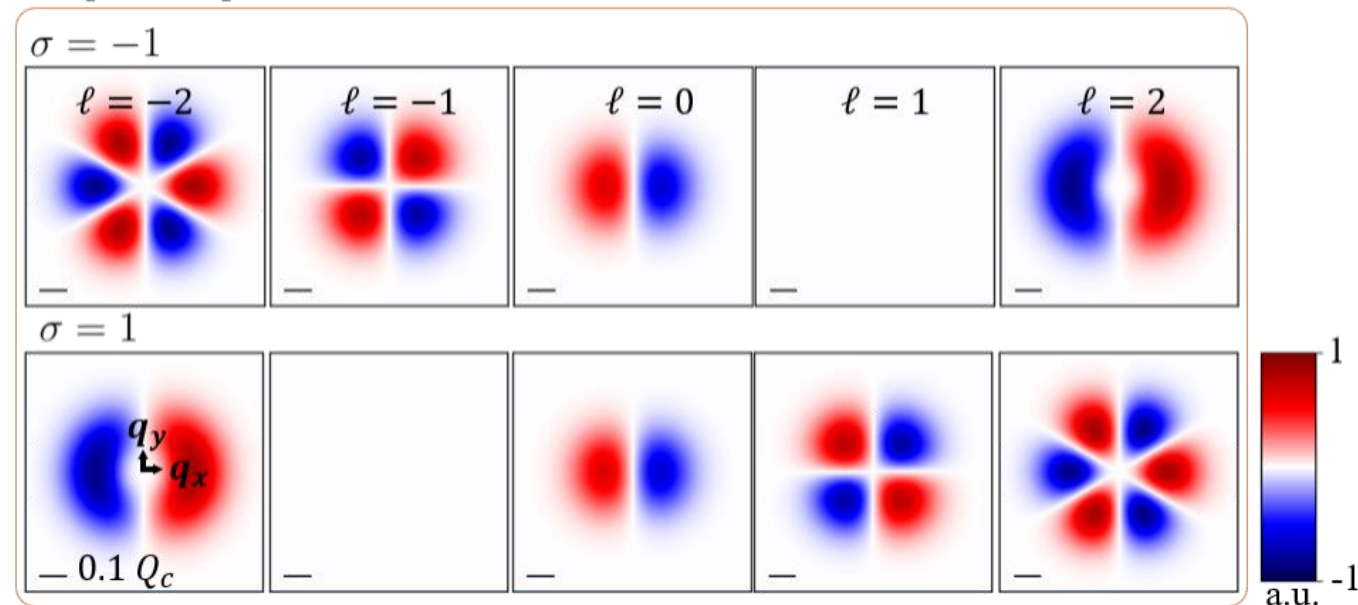
Transverse component :

$$\text{Re} [\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel})]$$



Longitudinal component :

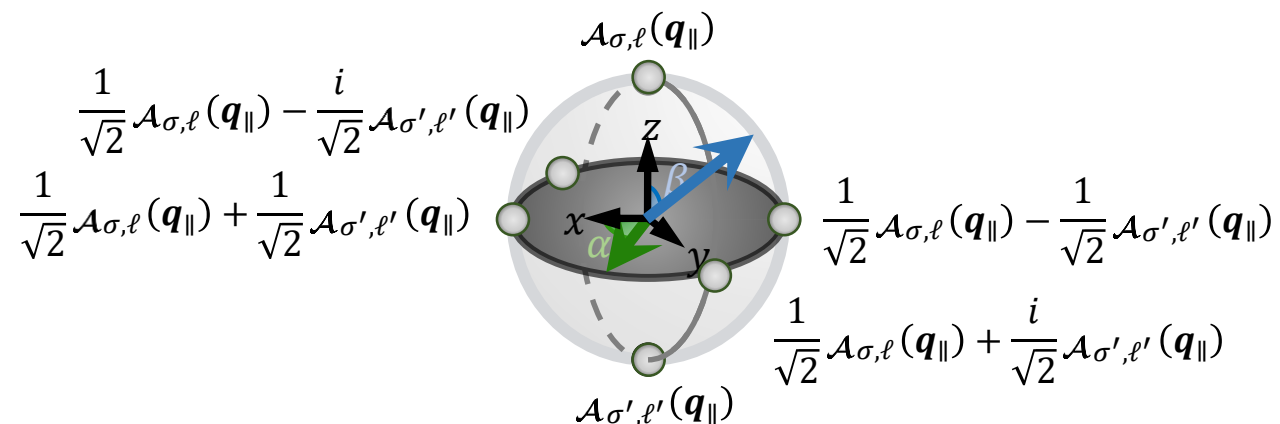
$$\text{Re} [\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel})]$$



2. Vector vortex beams (VVB)

Forming vector vortex beams (VVB) by Superimposing LG beams

$$\begin{aligned}\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) &= \cos\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma',\ell'}(\mathbf{q}_{\parallel}) \\ &= \tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) + \tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta)\end{aligned}$$



- Poincaré sphere for VVB states

2. VVB (Transverse component): $|\ell| = |\ell'|$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)$$

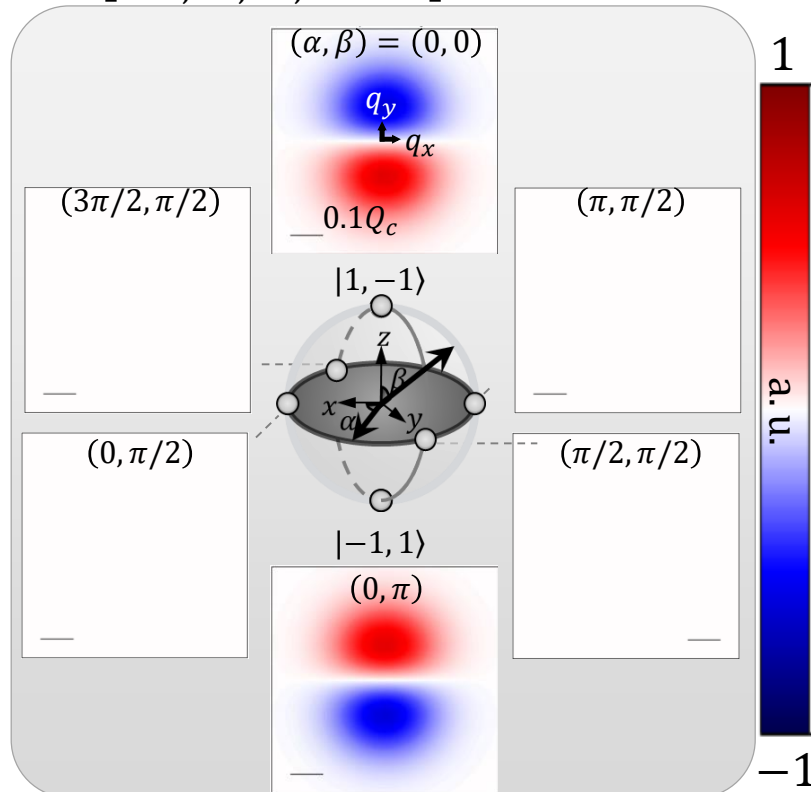
$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) = \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\bar{\ell}\phi_{q_{\parallel}}}$$

$$\hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{x}} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{y}} \right\}$$

Ex. $\ell = -\ell'$

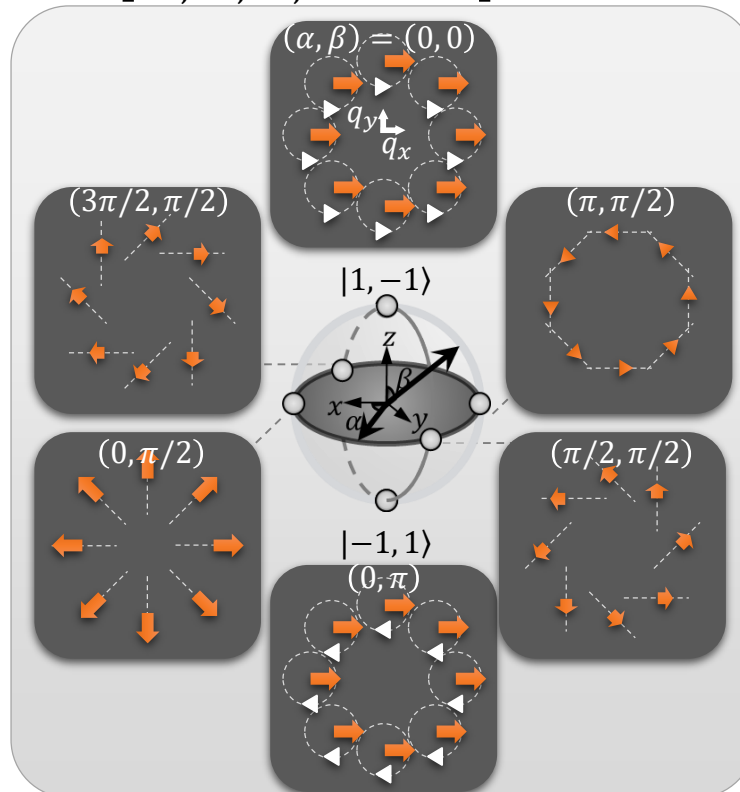
$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,1}^{\parallel}(\mathbf{q}_{\parallel})]$$

$$\bar{\ell} = 0$$



$$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,1}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)]$$

$$\Delta\ell = 2$$



Carrying phase : $\bar{\ell}\phi_{q_{\parallel}}$

$$\bar{\ell} \equiv \frac{\ell' + \ell}{2}$$

Polarization change : $\frac{\Delta\ell}{2}\phi_{q_{\parallel}}$

$$\Delta\ell \equiv \ell' - \ell$$

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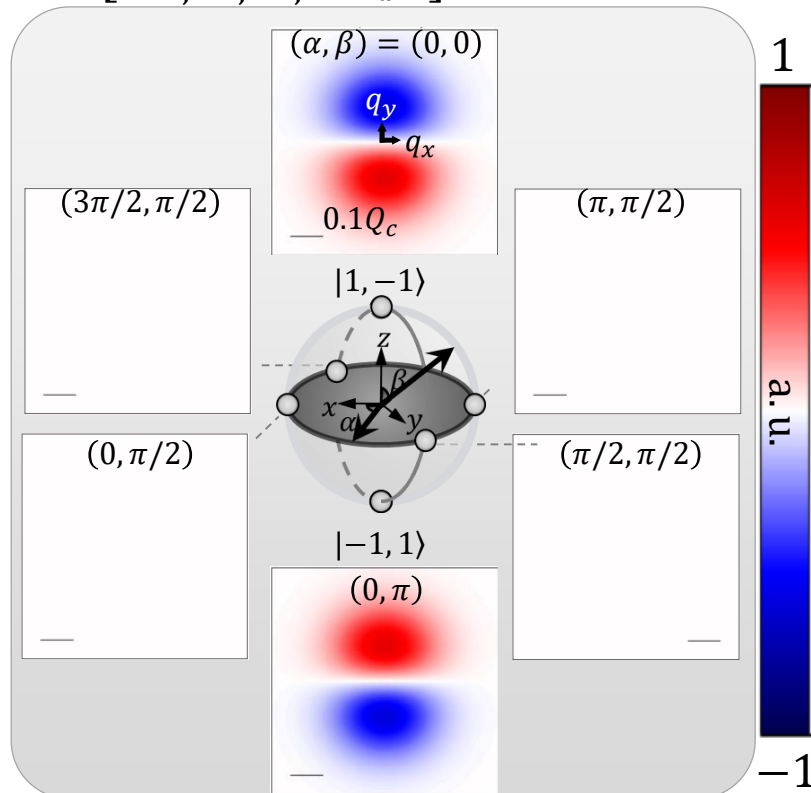
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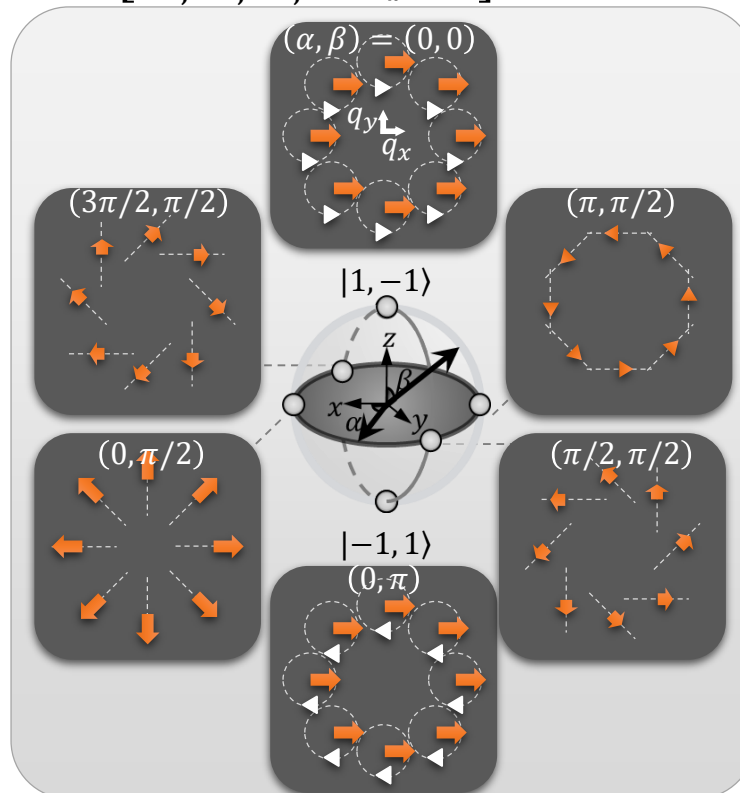
$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,1}^{\parallel}(\mathbf{q}_{\parallel})]$$

$$\bar{\ell} = 0$$

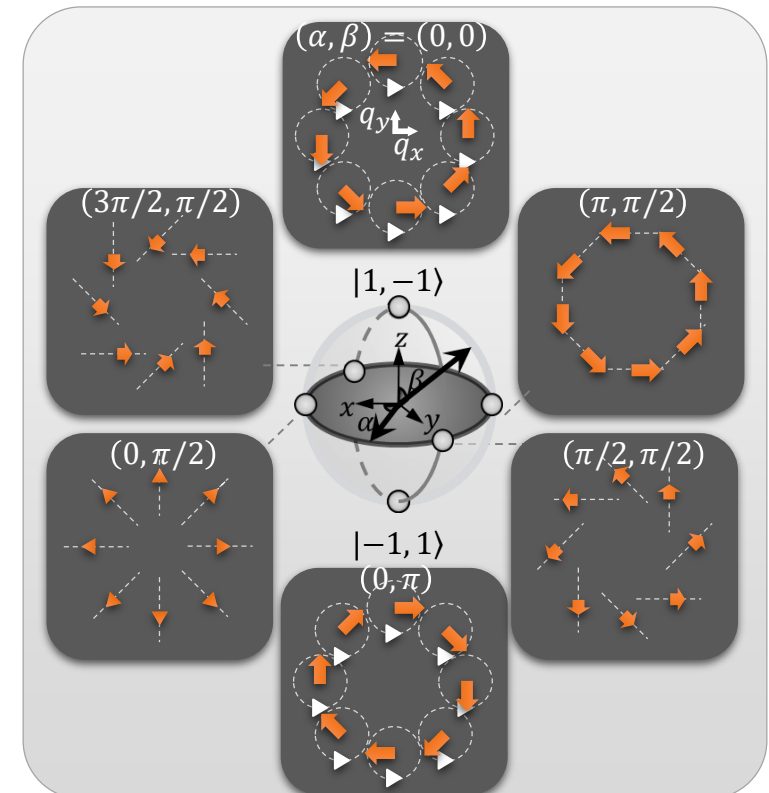


$$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,1}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)]$$

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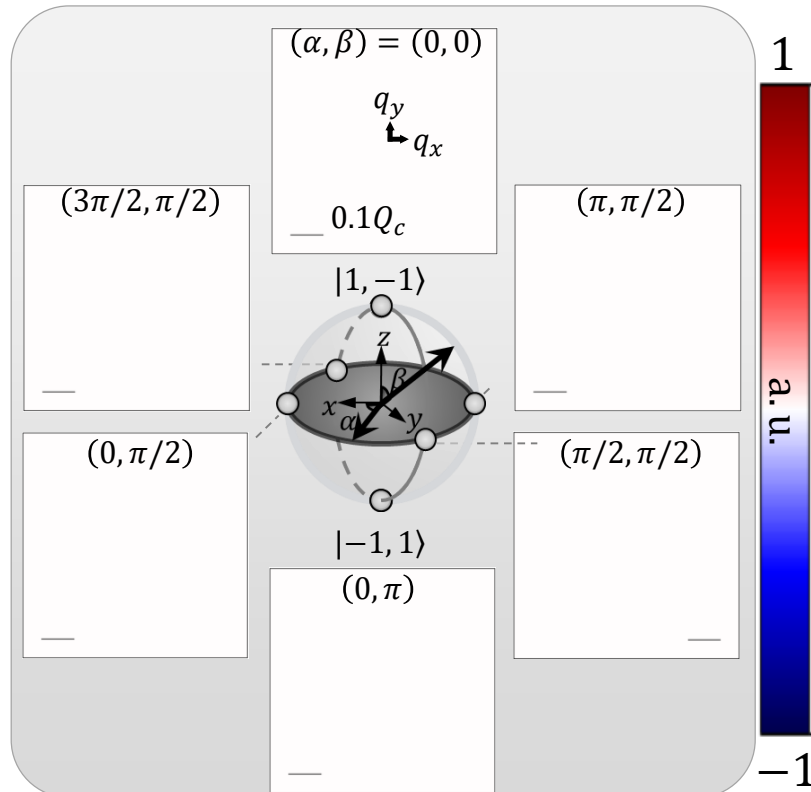
$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}) = \frac{\sin \theta_q}{2} \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\bar{J}\phi_{q_{\parallel}}}$$

$$\hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2} \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha}{2}\right)} \hat{\mathbf{z}}$$

Ex. $\ell = -\ell'$

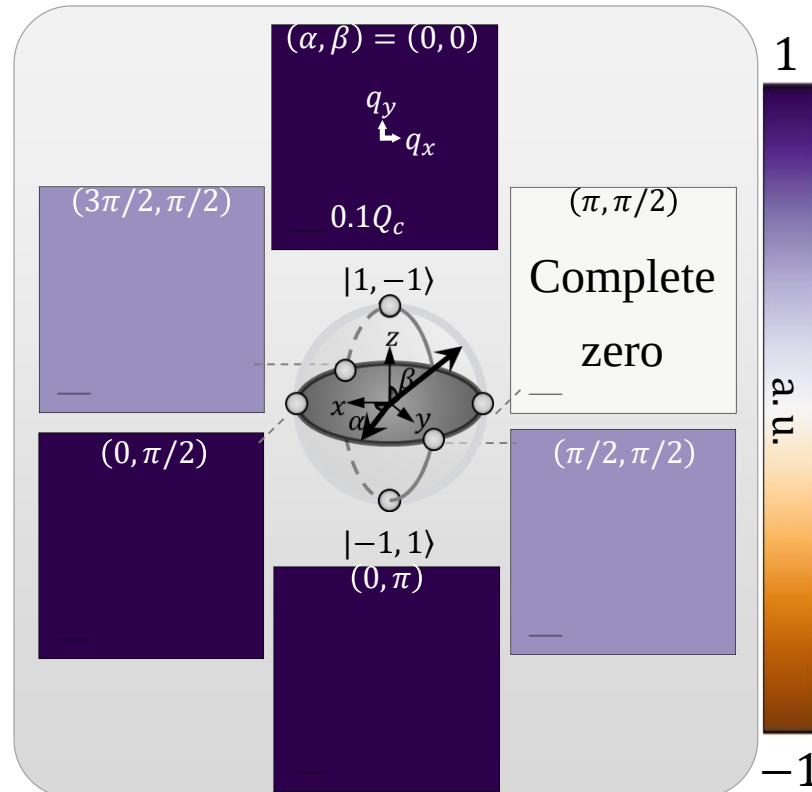
$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel})]$$

$$\bar{J} = 0$$



$$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel}; \alpha)]$$

$$\Delta J = 0$$



Carrying phase : $\bar{J}\phi_{q_{\parallel}}$

$$\bar{J} \equiv \frac{J' + J}{2}$$

Polarization change : $\frac{\Delta J}{2} \phi_{q_{\parallel}}$

$$\Delta J \equiv J' - J$$

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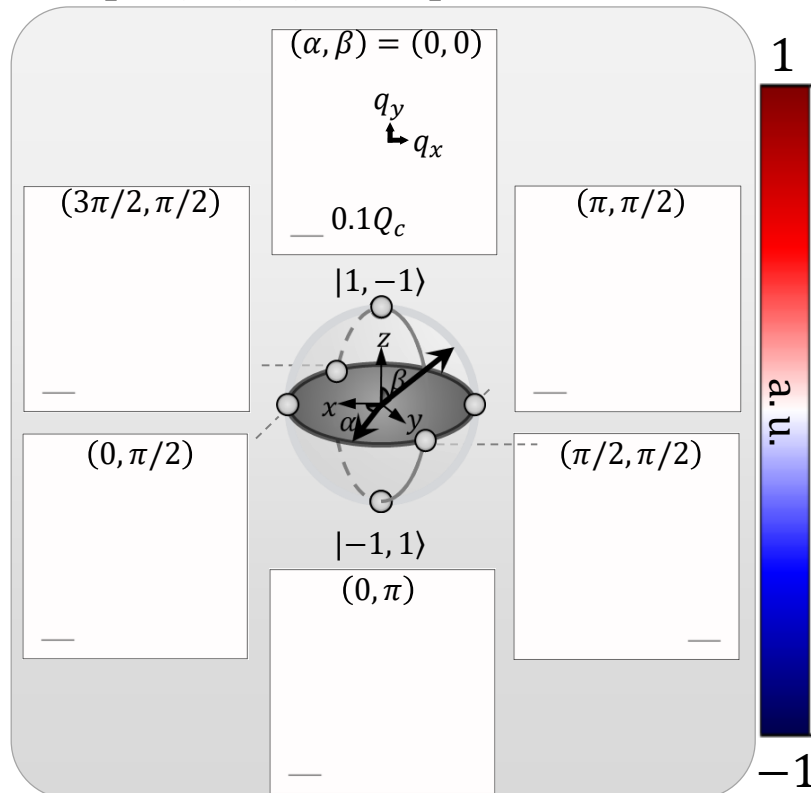
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Ex. $\ell = -\ell'$

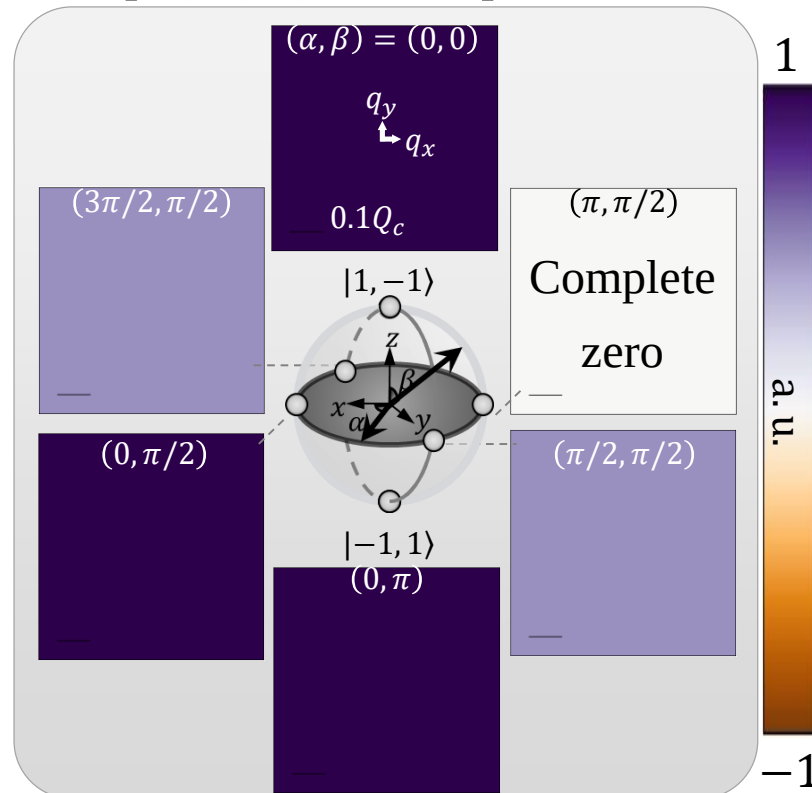
$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel})]$$

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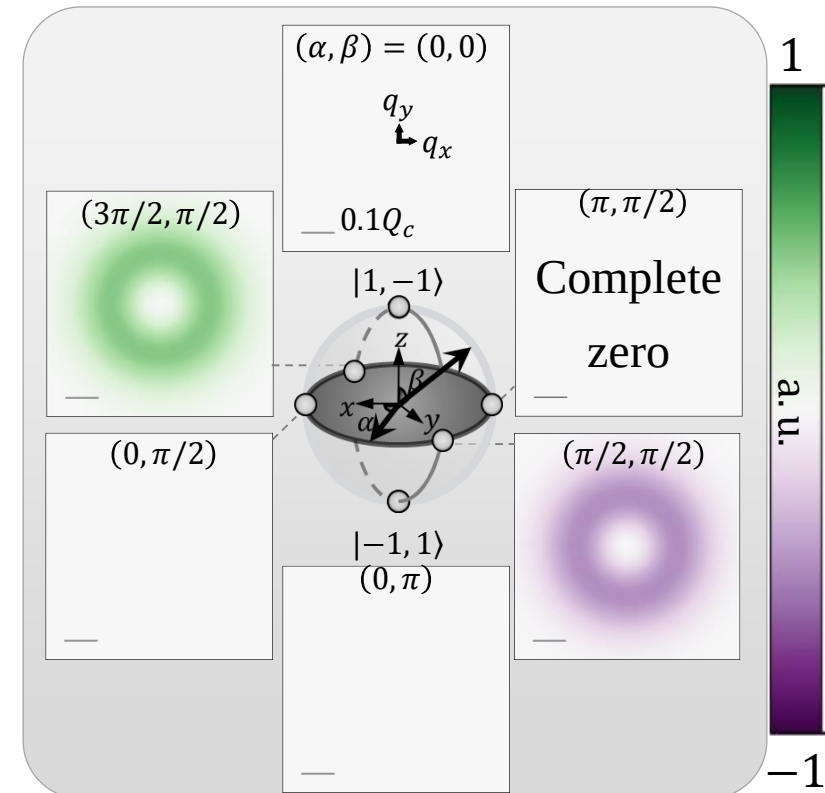


$$\text{Re}[\hat{\boldsymbol{\varepsilon}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel}; \alpha)]$$

$$\Delta J = 0$$



$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel}; \alpha, \beta)]$$

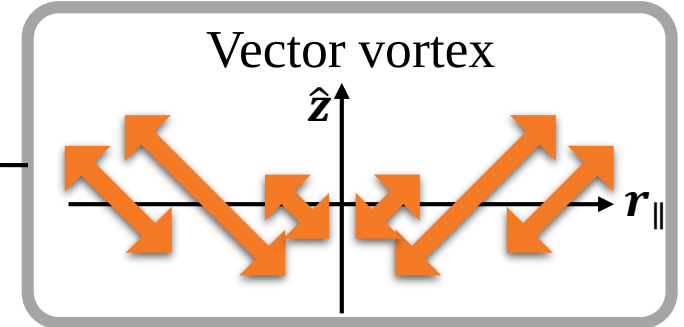


2. VVB : Brief overview

Summary for $\beta = \frac{\pi}{2}$, $\sigma = -\sigma' = 1$, and $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)$$

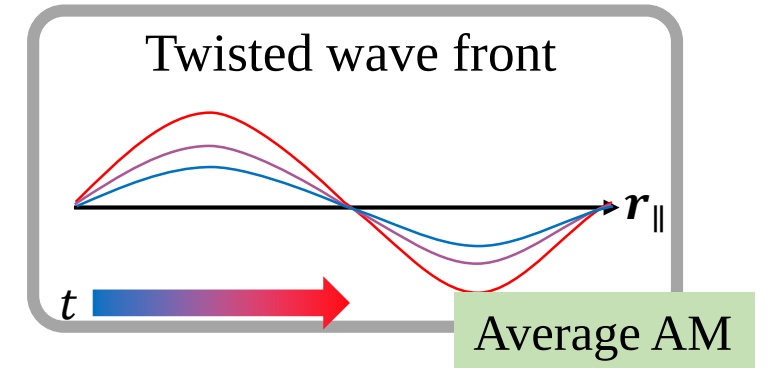
$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha)$$



Transverse component:

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{\ell}\phi_{q_{\parallel}}}$$

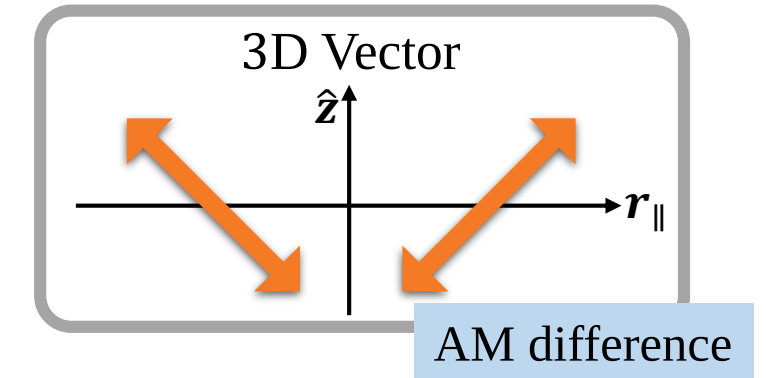
$$\hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{x}} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{y}} \right\}$$



Longitudinal component:

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}) = -\frac{\sin\theta_q}{2} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{J}\phi_{q_{\parallel}}}$$

$$\hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{z}}$$



2. VVB : Transverse component

Relax the condition $|\ell| \neq |\ell'|$

$$\begin{aligned} \tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) &= e^{i\bar{\ell}\phi_{q_{\parallel}}} \times \frac{1}{2} e^{i(-\frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{\mathbf{x}} \right. \\ &\quad \left. + \left[\sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) + i \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{\mathbf{y}} \right\} \end{aligned}$$

1. Define $G_{|\ell|,|\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})]$

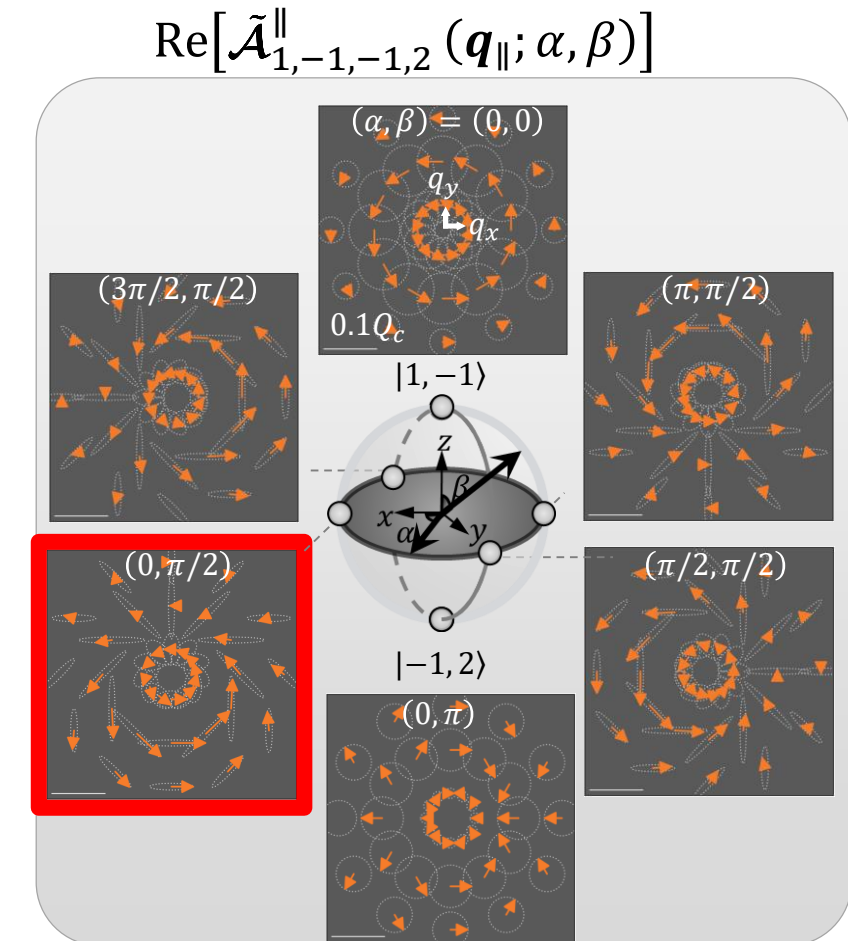
$$H_{|\ell|,|\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})]$$

$$\Delta|\ell| \equiv |\ell'| - |\ell|$$

$$|\bar{\ell}| \equiv (|\ell'| + |\ell|)/2$$

2. Asymmetry polarization : $\Delta\ell = 3$

3. Asymmetry phase : $\bar{\ell} = \frac{1}{2}$



2. VVB : Transverse component

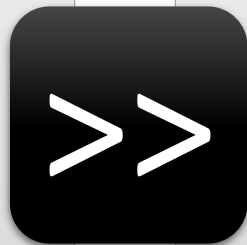
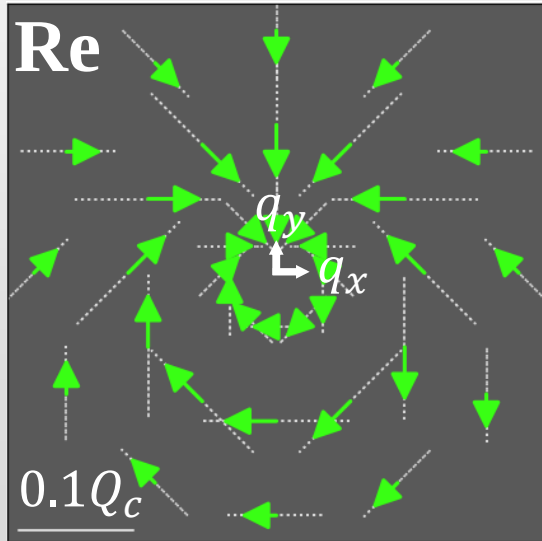
Relax the condition $|\ell| \neq |\ell'|$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = e^{i\bar{\ell}\phi_{q_{\parallel}}} \times \frac{1}{2} e^{i(-\frac{\pi}{2}|\ell| + \frac{\alpha}{2})} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{x} \right. \\ \left. + \left[\sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) + i \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{y} \right\}$$

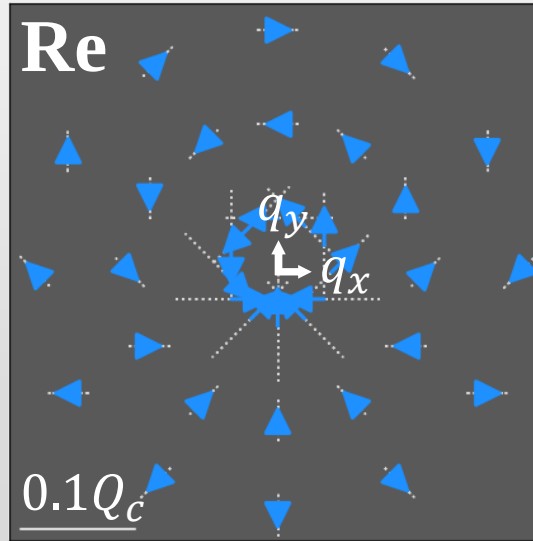
$$\propto G_{|-1|,|2|}(q_{\parallel})$$

$$\propto 3 \times H_{|-1|,|2|}(q_{\parallel})$$

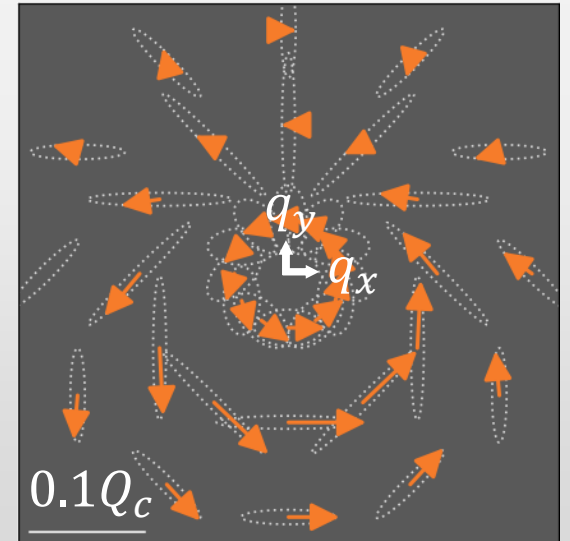
Asymmetry polarization $\Delta\ell = 3$



Asymmetry polarization $\Delta\ell = 3$



$$\text{Re} \left[\tilde{\mathcal{A}}_{1,-1,-1,2}^{\parallel}(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \frac{\pi}{2}) \right]$$



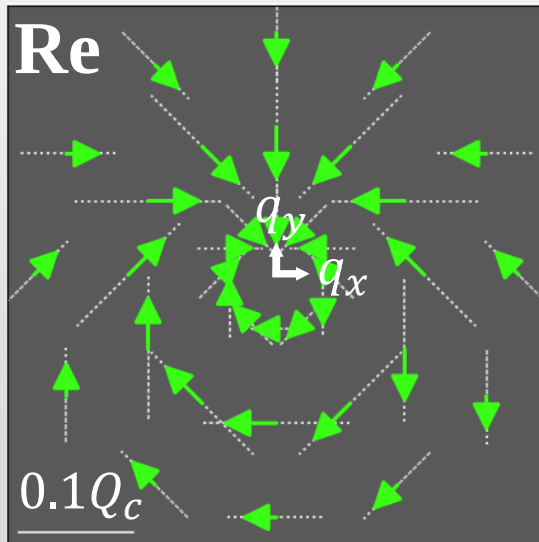
2. VVB : Transverse component

Relax the condition $|\ell| \neq |\ell'|$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = e^{i\bar{\ell}\phi_{q_{\parallel}}} \times \frac{1}{2} e^{i(-\frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{\mathbf{x}} + \left[\sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) + i \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{\mathbf{y}} \right\}$$

$$\propto G_{|-1|,|2|}(q_{\parallel})$$

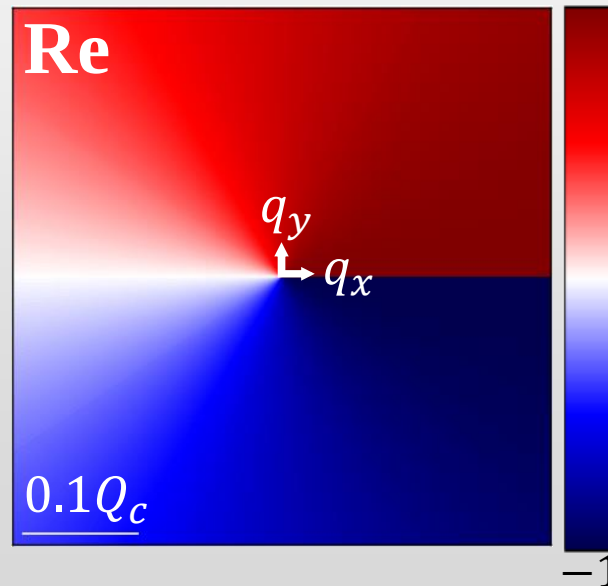
Asymmetry polarization $\Delta\ell = 3$



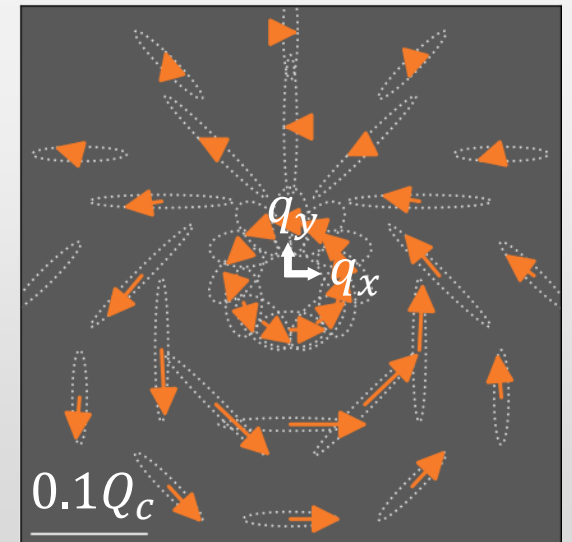
$$e^{i\bar{\ell}\phi_{q_{\parallel}}}$$

Asymmetry phase

$$\bar{\ell} = \frac{1}{2}$$



$$\text{Re} \left[\tilde{\mathcal{A}}_{1,-1,-1,2}^{\parallel}(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \frac{\pi}{2}) \right]$$



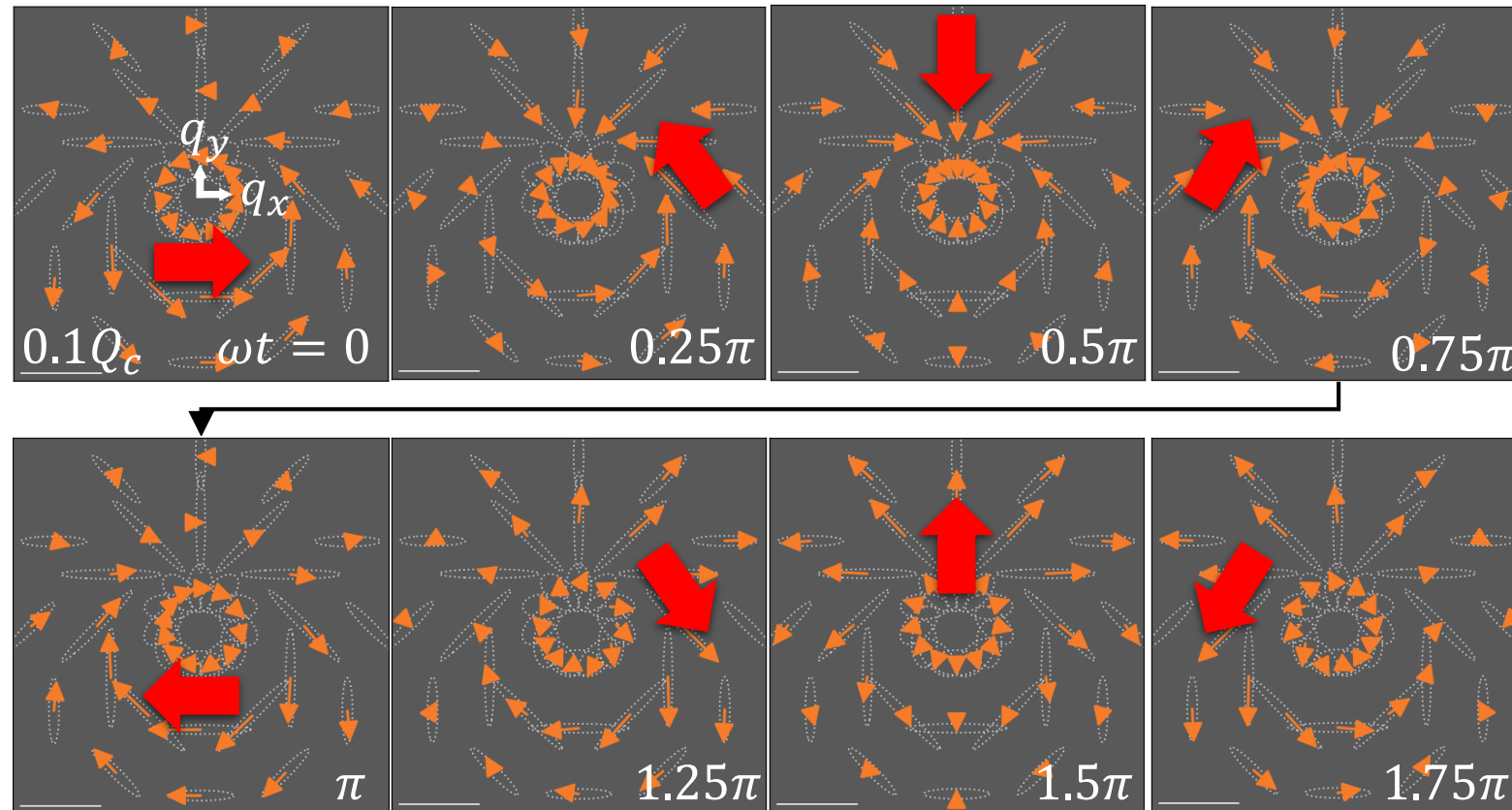
2. VVB : Transverse component

Relax the condition $|\ell| \neq |\ell'|$

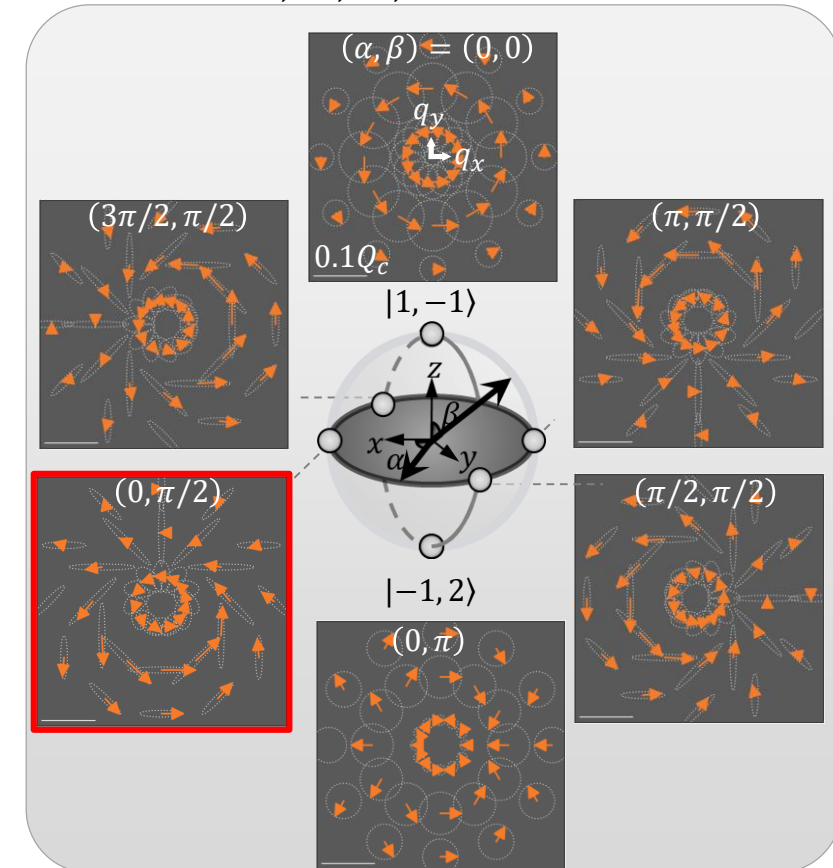
$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} = e^{i(\bar{\ell}\phi_{q_{\parallel}} - \omega t)} \frac{1}{2} e^{i(-\frac{\pi}{2}|\ell| + \frac{\alpha}{2})} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{\mathbf{x}} \right. \\ \left. + \left[\sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) + i \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right] \hat{\mathbf{y}} \right\}$$

$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,2}^{\parallel}(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \pi/2) e^{-i\omega t}]$$

Main component



$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,2}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta)]$$



2. VVB : Longitudinal component

Relax the condition $|\ell| \neq |\ell'|$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = e^{i\bar{J}\phi_{q_{\parallel}}} \times -\frac{\sin \theta_{\mathbf{q}}}{2} \cdot e^{i(-\frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

1. With

$$G_{|\ell|,|\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})]$$

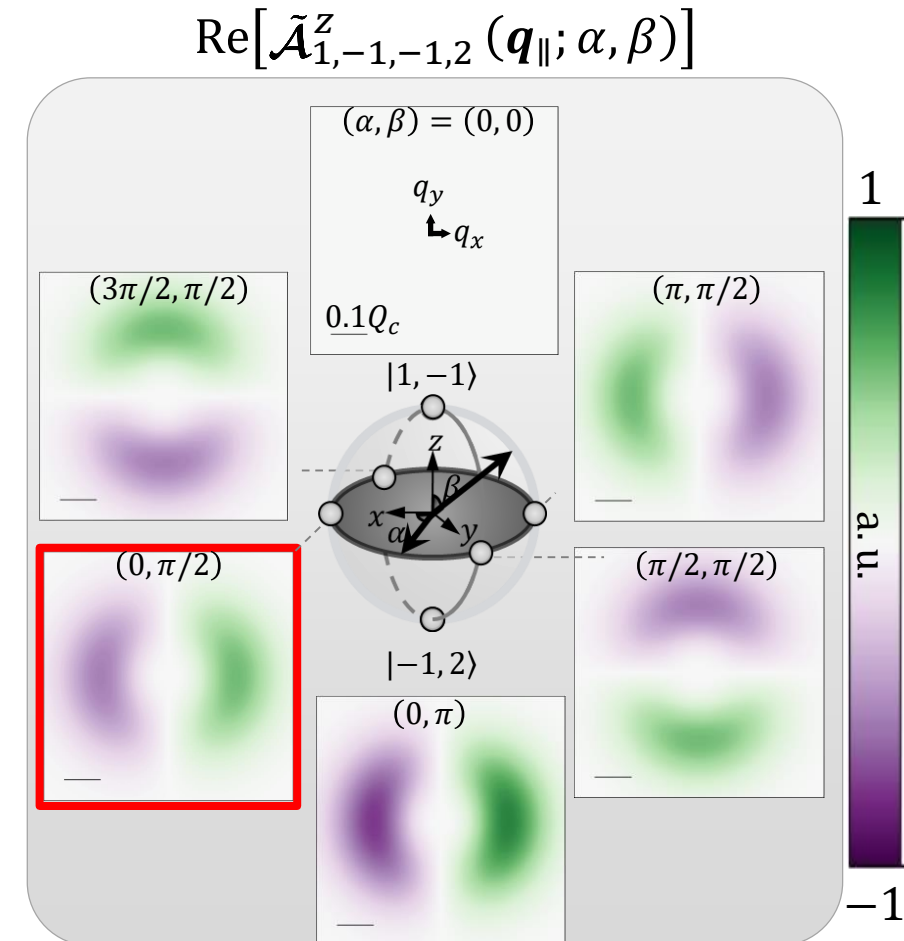
$$H_{|\ell|,|\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})]$$

$$\Delta|\ell| \equiv |\ell'| - |\ell|$$

$$|\bar{\ell}| \equiv (|\ell'| + |\ell|)/2$$

2. Asymmetry pattern : $\Delta J = 1$

3. Asymmetry phase : $\bar{J} = \frac{1}{2}$



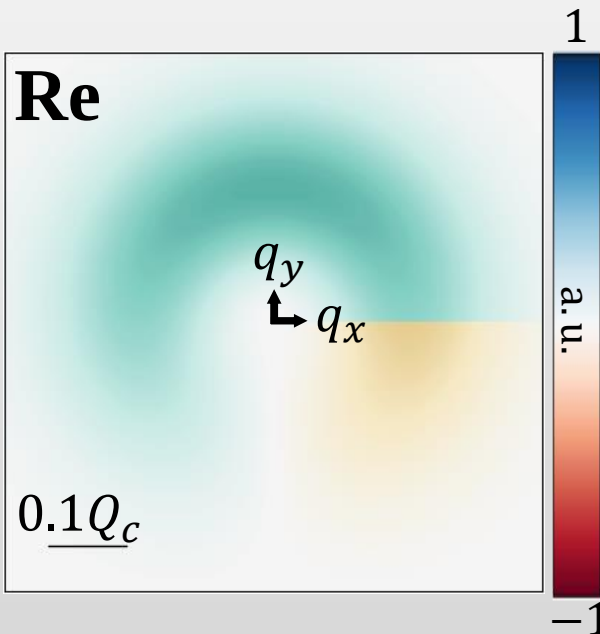
2. VVB : Longitudinal component

Relax the condition $|\ell| \neq |\ell'|$

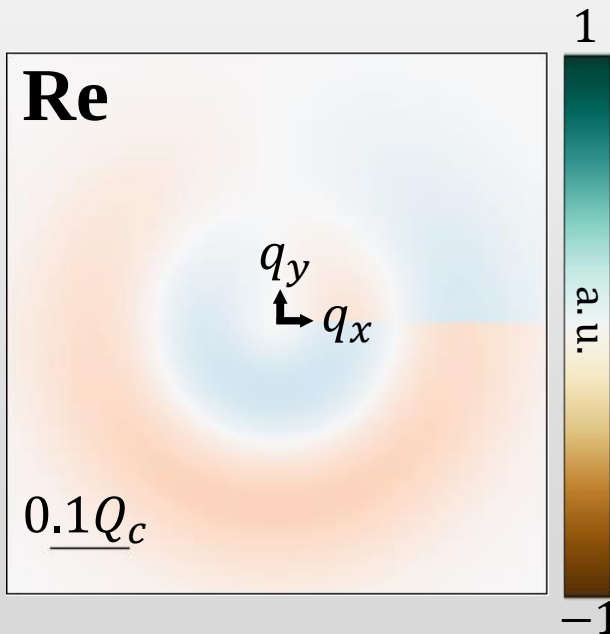
$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = e^{i\bar{J}\phi_{q_{\parallel}}} \times -\frac{\sin \theta_{\mathbf{q}}}{2} \cdot e^{i(-\frac{\pi}{2}|\ell| + \frac{\alpha}{2})} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

$\propto G_{|-1|,|2|}(q_{\parallel})$

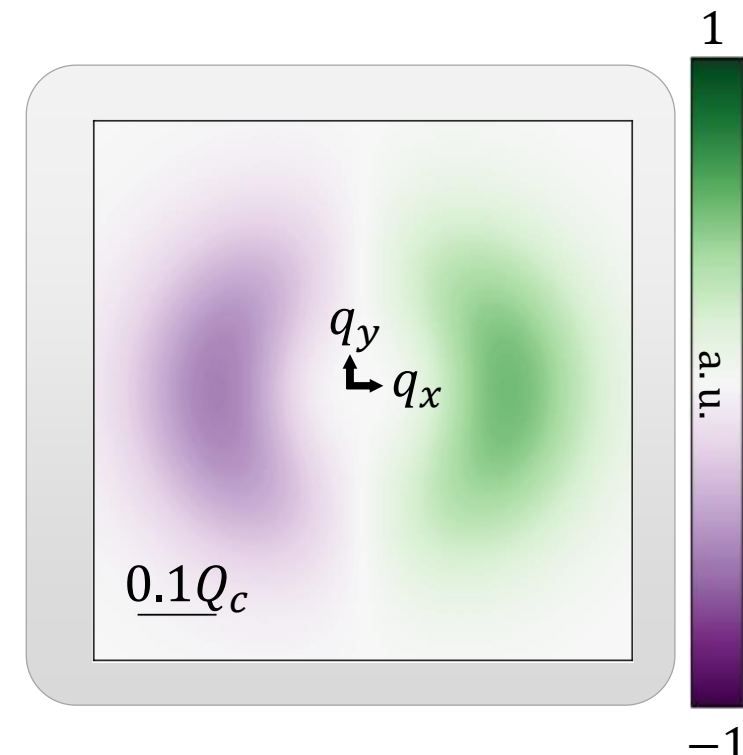
Asymmetry pattern $\Delta J = 1$



$\propto 3 \times H_{|-1|,|2|}(q_{\parallel})$



$\text{Re} \left[\tilde{\mathcal{A}}_{1,-1,-1,2}^z(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \frac{\pi}{2}) \right]$



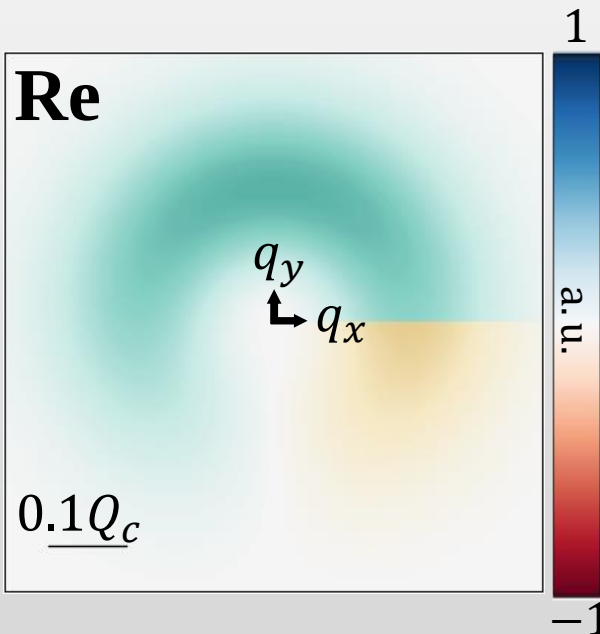
2. VVB : Longitudinal component

Relax the condition $|\ell| \neq |\ell'|$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = e^{i\bar{J}\phi_{q_{\parallel}}} \times -\frac{\sin \theta_{\mathbf{q}}}{2} \cdot e^{i(-\frac{\pi}{2}|\ell| + \frac{\alpha}{2})} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

$\propto G_{|-1|,|2|}(q_{\parallel})$

Asymmetry pattern $\Delta J = 1$

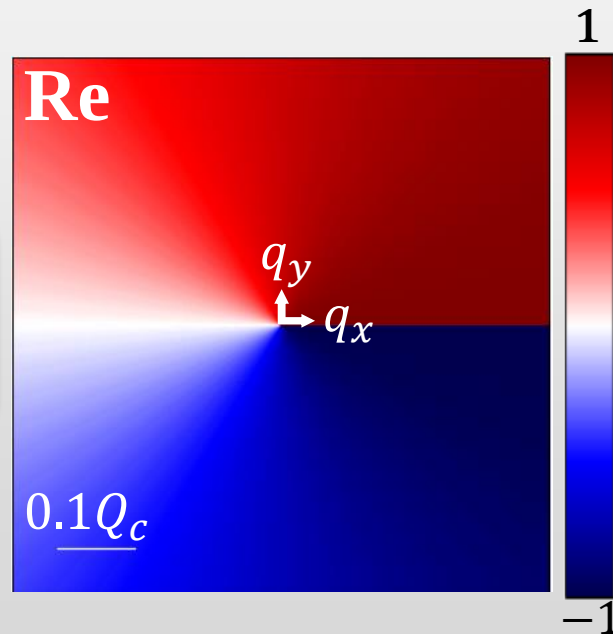


both
asymmetry

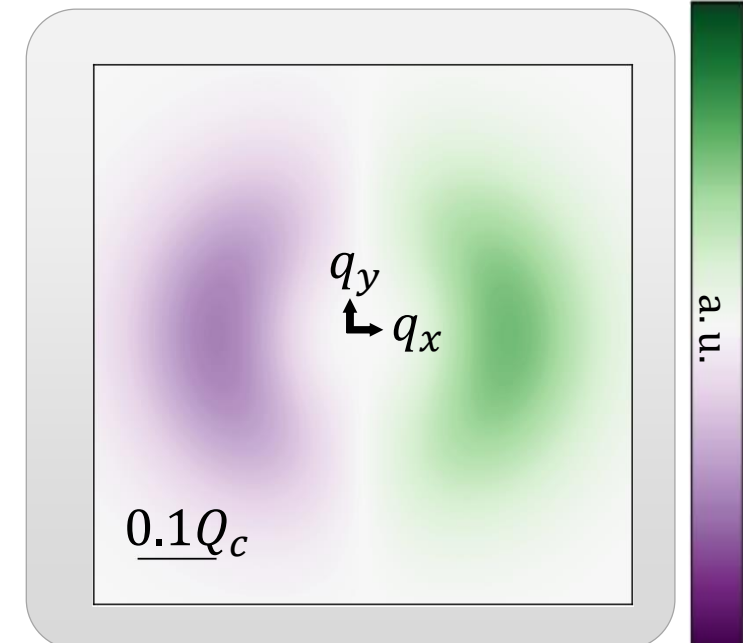
$e^{i\bar{J}\phi_{q_{\parallel}}}$

Asymmetry phase

$\bar{J} = \frac{1}{2}$



$\text{Re} \left[\tilde{\mathcal{A}}_{1,-1,-1,2}^z(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \frac{\pi}{2}) \right]$

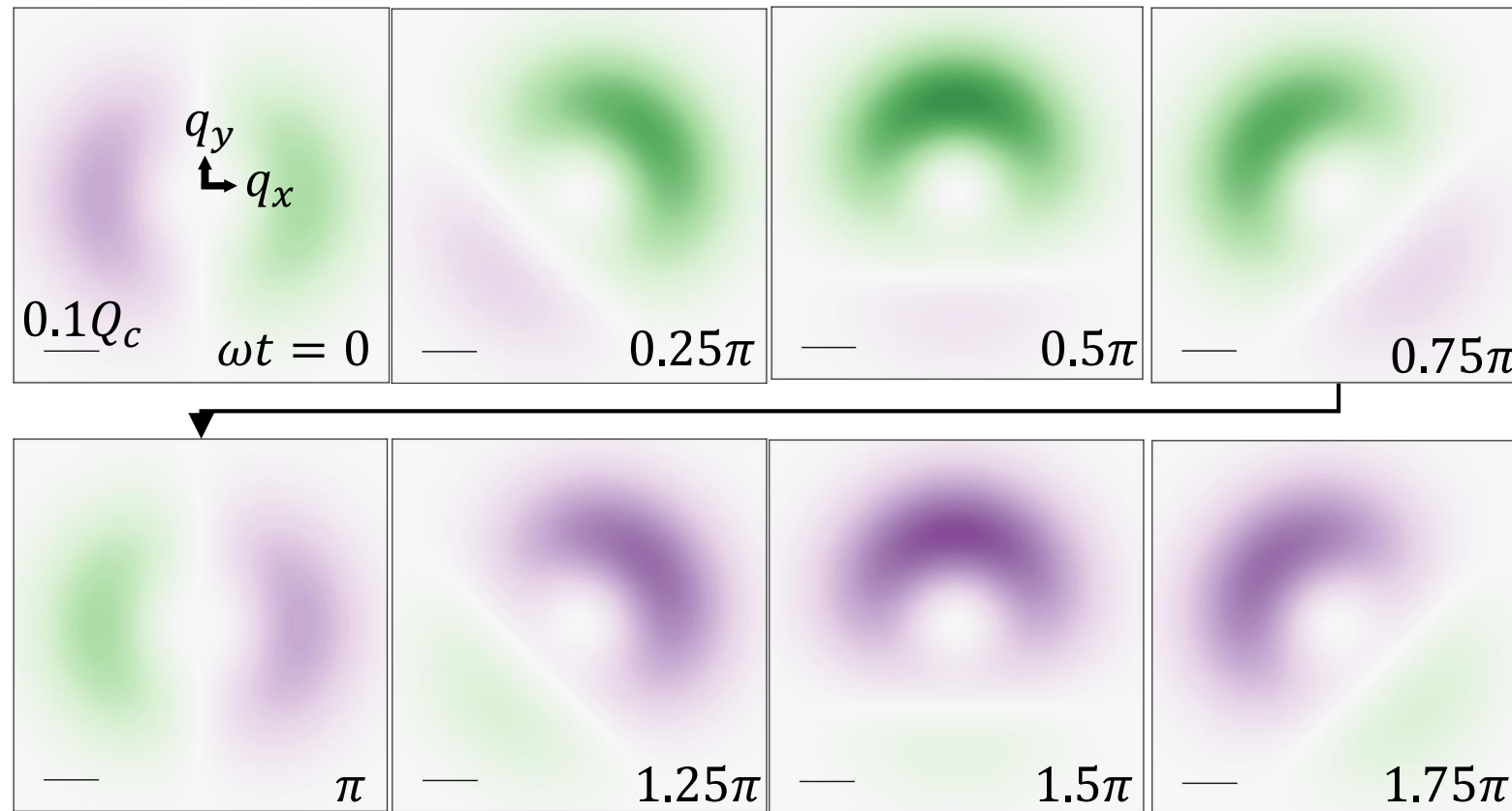


2. VVB : Longitudinal component

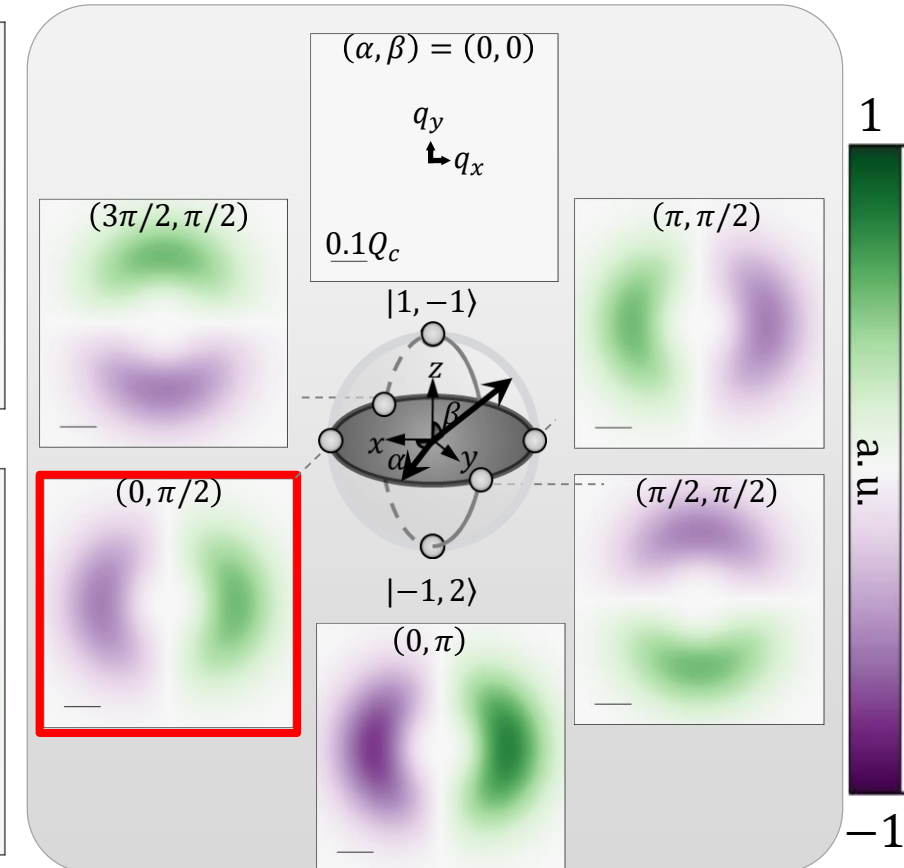
Relax the condition $|\ell| \neq |\ell'|$

$$\tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} = e^{i(\bar{J}\phi_{q_{\parallel}} - \omega t)} \times -\frac{\sin \theta_q}{2} \cdot e^{i(-\frac{\pi}{2}|\ell| + \frac{\alpha}{2})} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,2}^z(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \pi/2) e^{-i\omega t}]$$



$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,2}^z(\mathbf{q}_{\parallel}; \alpha, \beta)]$$



1. Introduction

2. Spin-orbital angular momentum interaction in light

3. Light-matter interaction

- Fell selective excitation
- Directional selective excitation

4. Exciton wave packet

Summary and outlook

Light matter interaction

Following O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024), the transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha, \beta) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 \rho(\hbar\omega = E_{S,Q}^X)$$

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha, \beta) \propto \mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha, \beta) \cdot \mathbf{D}^{S,Q}$$

The conditions $\sigma = -\sigma' = 1$, $\beta = \frac{\pi}{2}$, and $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$ give **optical matrix elements** ($\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha, \beta)$) as

$$\tilde{M}_{1,\ell,-1,\ell'}^{B\pm,Q}(\mathbf{Q}; \alpha, \beta = \frac{\pi}{2}) \propto \tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) (\hat{\boldsymbol{\varepsilon}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) \cdot \mathbf{D}^{B\pm,Q})$$

$$\tilde{M}_{1,\ell,-1,\ell'}^{G,Q}(\mathbf{Q}; \alpha, \beta = \frac{\pi}{2}) \propto \tilde{\mathcal{A}}_{1,\ell,-1,\ell'}^Z(\mathbf{q}_{\parallel}) (\hat{\boldsymbol{\varepsilon}}_{1,\ell,-1,\ell'}^Z(\mathbf{q}_{\parallel}; \alpha) \cdot \mathbf{D}^{G,Q})$$

$$\hat{\boldsymbol{\varepsilon}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{x}} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{y}} \right\}$$

$$\hat{\boldsymbol{\varepsilon}}_{1,\ell,-1,\ell'}^Z(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{z}}$$

$\times$

$$\mathbf{D}^{B+,Q} = D^B (\cos \phi_Q \hat{\mathbf{x}} + \sin \phi_Q \hat{\mathbf{y}}) / \sqrt{2}$$

$$\mathbf{D}^{B-,Q} = iD^B (-\sin \phi_Q \hat{\mathbf{x}} + \cos \phi_Q \hat{\mathbf{y}}) / \sqrt{2}$$

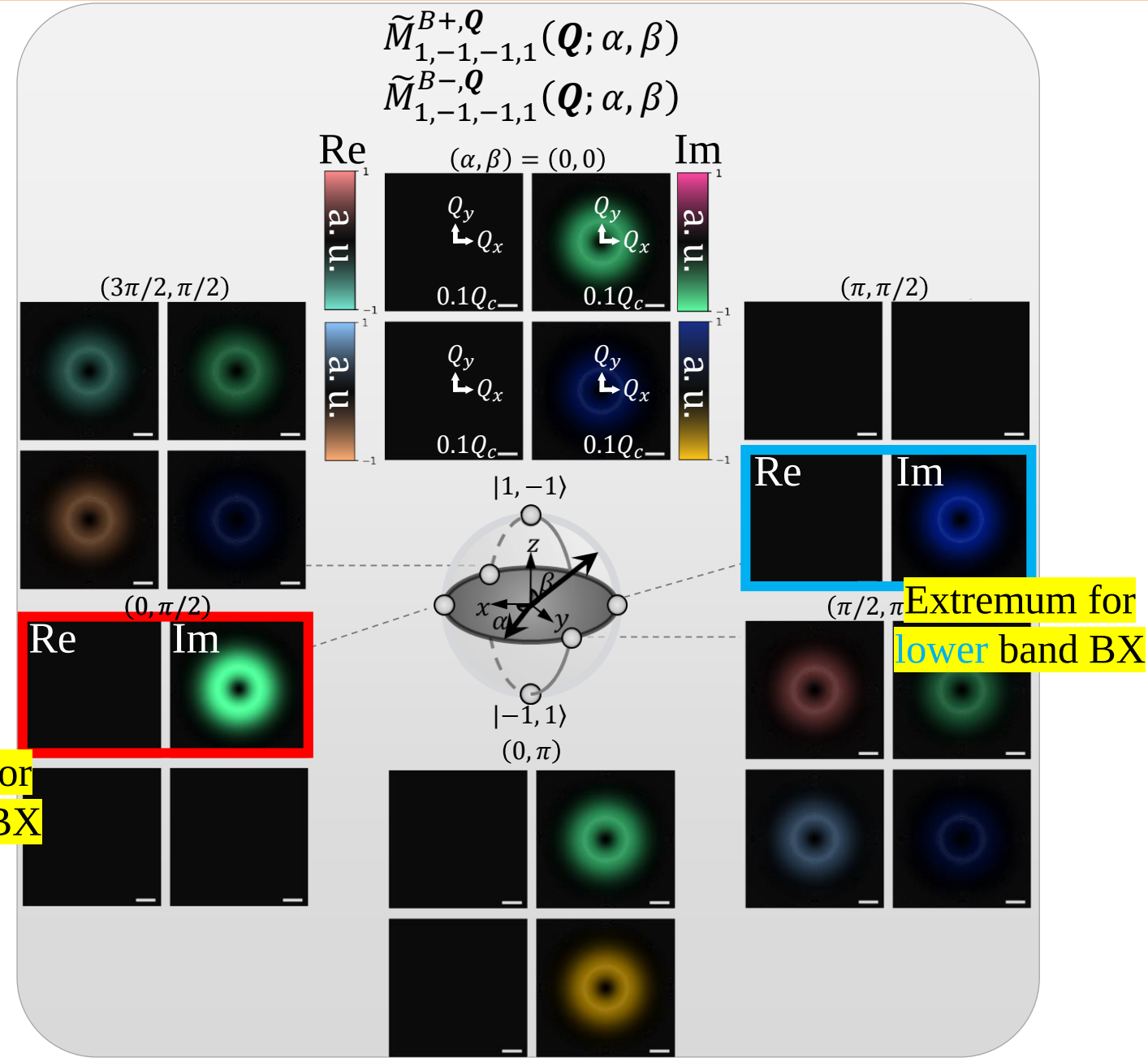
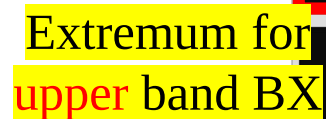
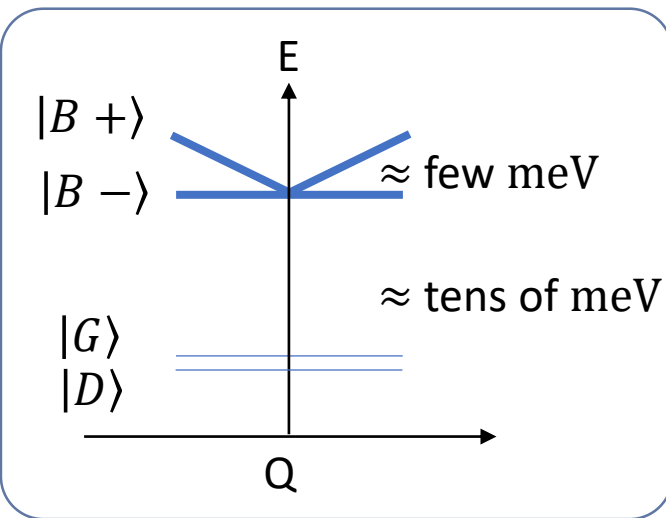
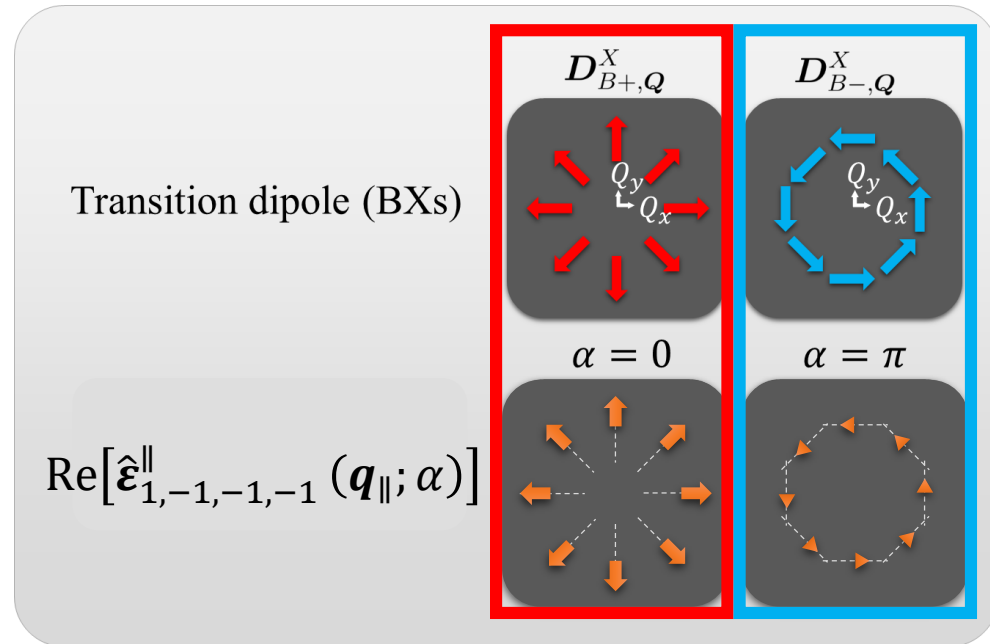
$$\mathbf{D}^{G,Q} = D^G \hat{\mathbf{z}}$$

Full selective excitation

With the conditions

Full selective excitation
only happened when $\ell = -1$

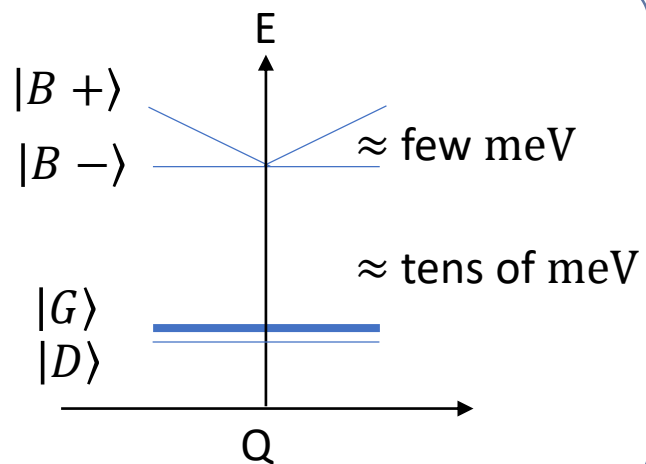
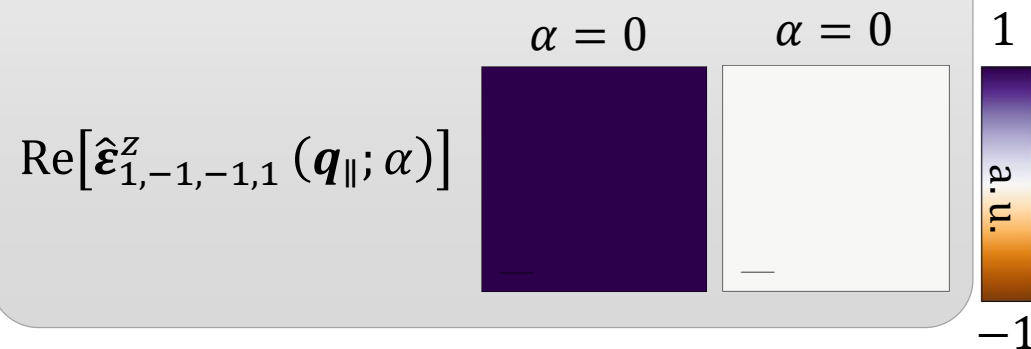
Full selective excitation



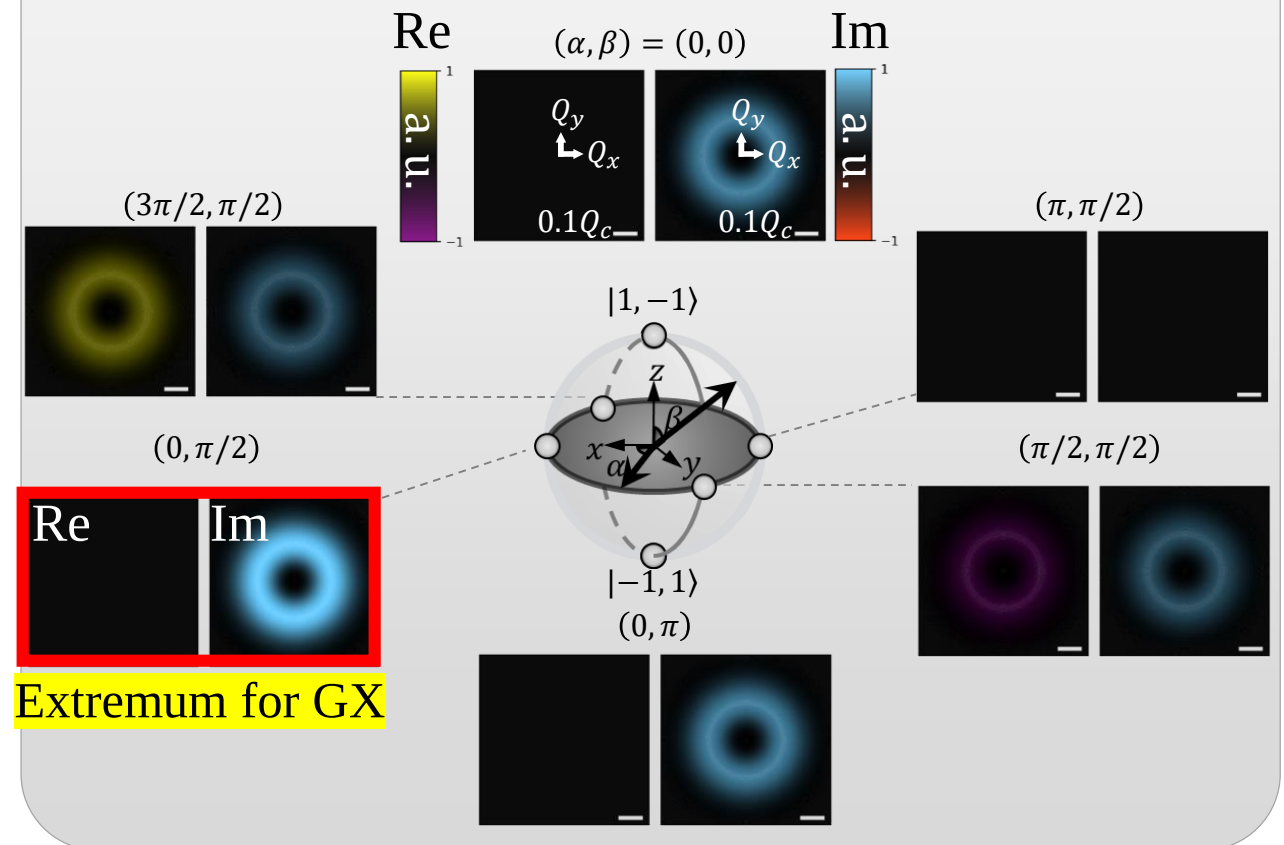
Full selective excitation

Transition dipole (GX)

$$\mathbf{D}^{G,Q} = D^G \hat{\mathbf{z}}$$



$$\tilde{M}_{1,-1,-1,1}^{G,Q}(\mathbf{Q}; \alpha, \beta)$$



Total Transition Rate

Summation over the transition rate

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^S(\alpha, \beta) = \sum_{\mathbf{Q}} \Gamma_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}; \alpha, \beta)$$

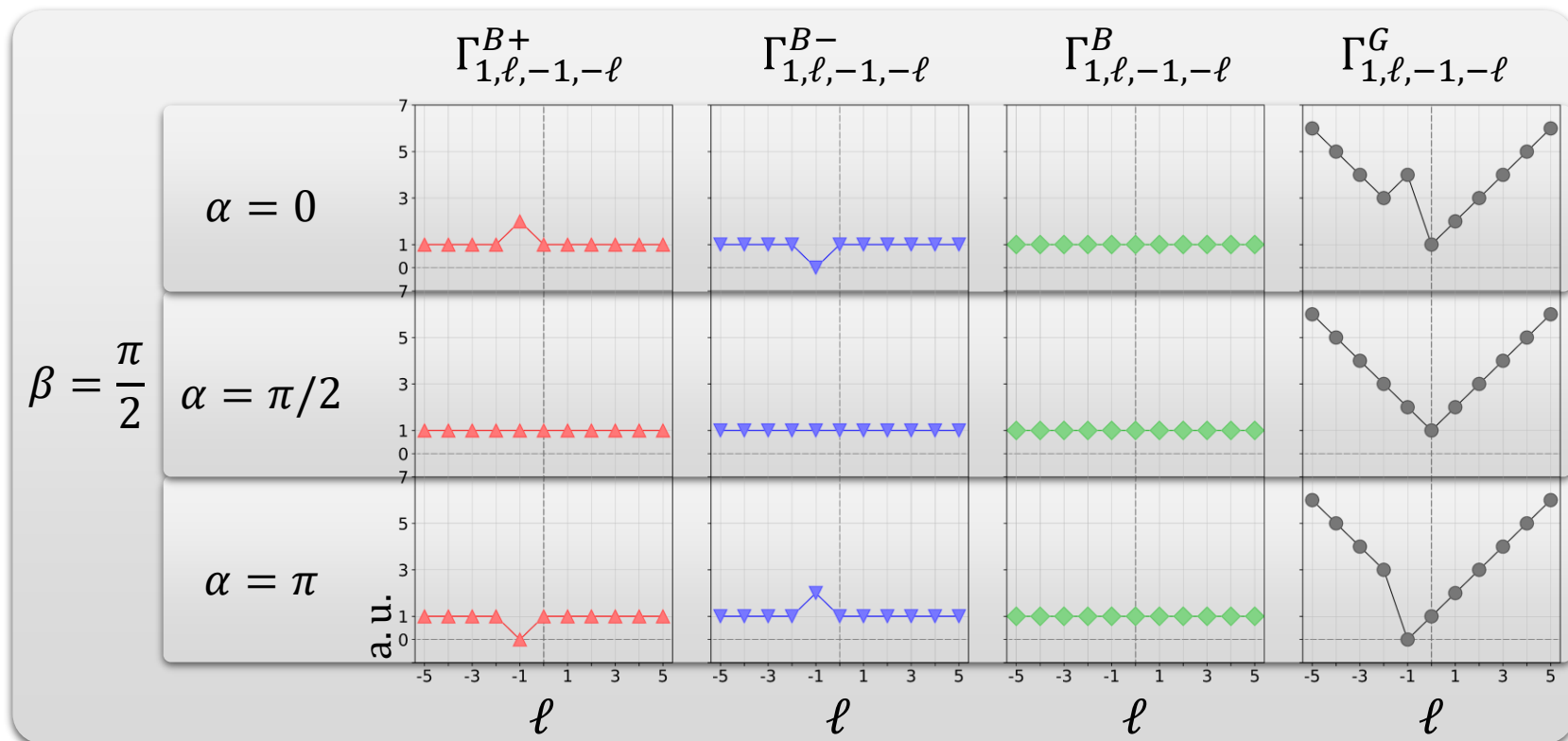
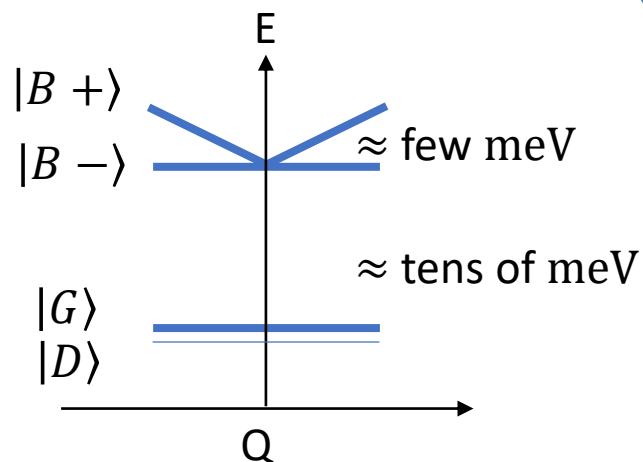
$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}; \alpha, \beta) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}; \alpha, \beta) \right|^2 \rho(\hbar\omega = E_{S,\mathbf{Q}}^X)$$

With the conditions

Full selective excitation
only happened when $\ell = -1$

Side note

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^B(\alpha, \beta) = \sum_{B^\pm} \Gamma_{\sigma,\ell,\sigma',\ell'}^{B^\pm}(\alpha, \beta)$$



Total Transition Rate

Summation over the whole plane

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^S(\alpha, \beta) = \sum_{\mathbf{Q}} \Gamma_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}; \alpha, \beta)$$

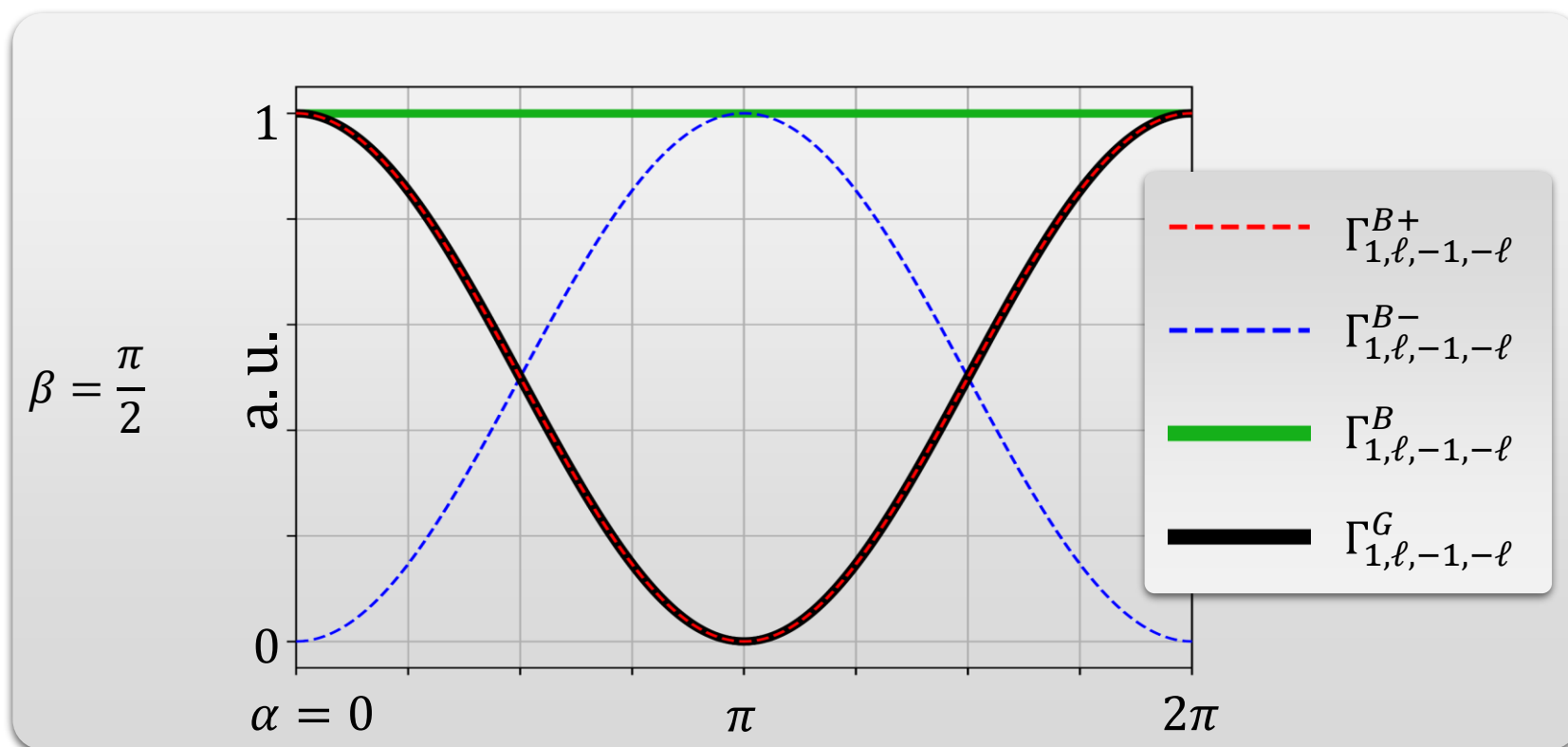
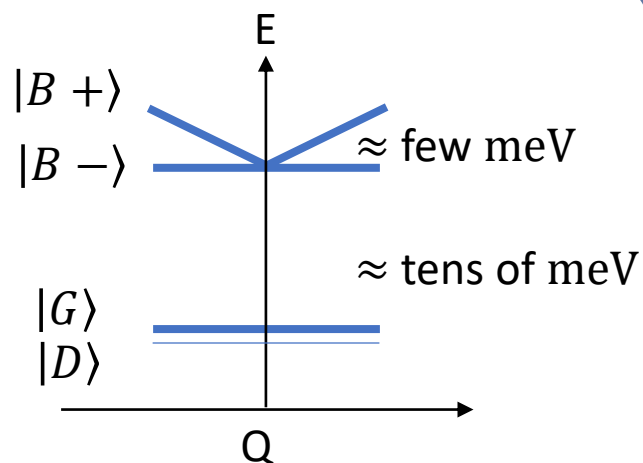
$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}; \alpha, \beta) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}; \alpha, \beta) \right|^2 \rho(\hbar\omega = E_{S,\mathbf{Q}}^X)$$

With the conditions

Full selective excitation
changes with α , $\beta = \frac{\pi}{2}$

Side note

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^B(\alpha, \beta) = \sum_{B\pm} \Gamma_{\sigma,\ell,\sigma',\ell'}^{B\pm}(\alpha, \beta)$$

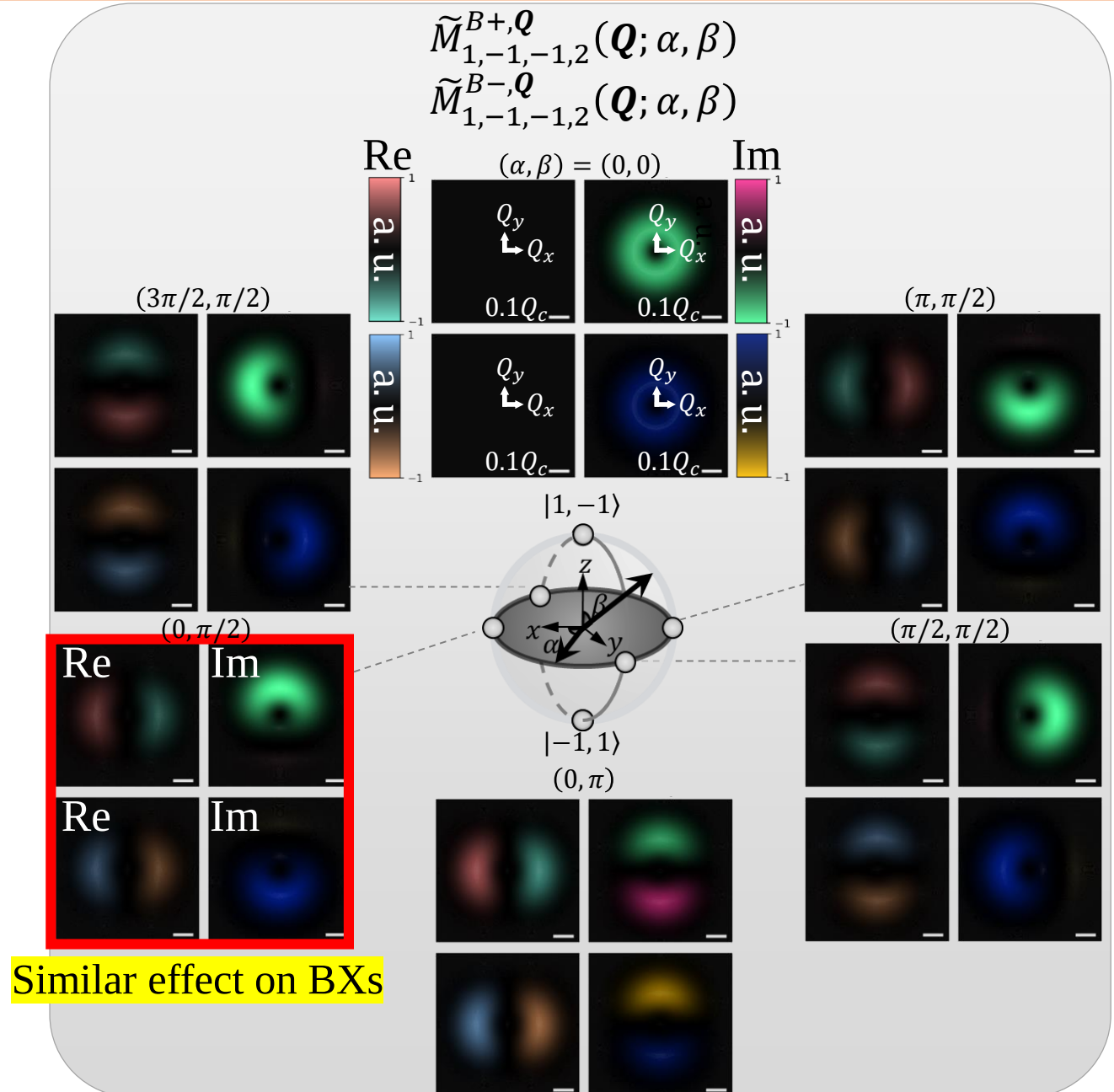
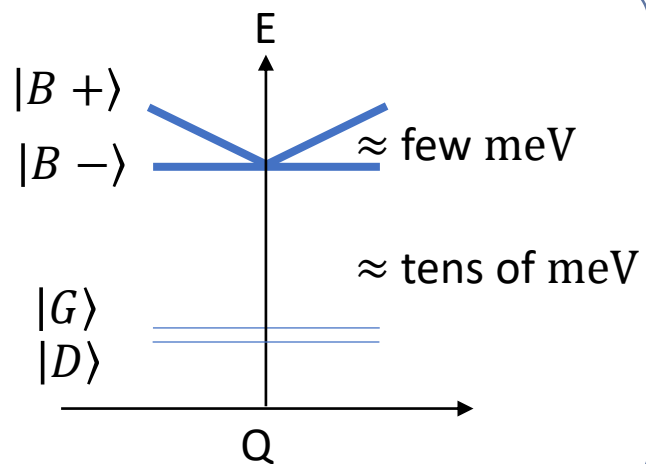
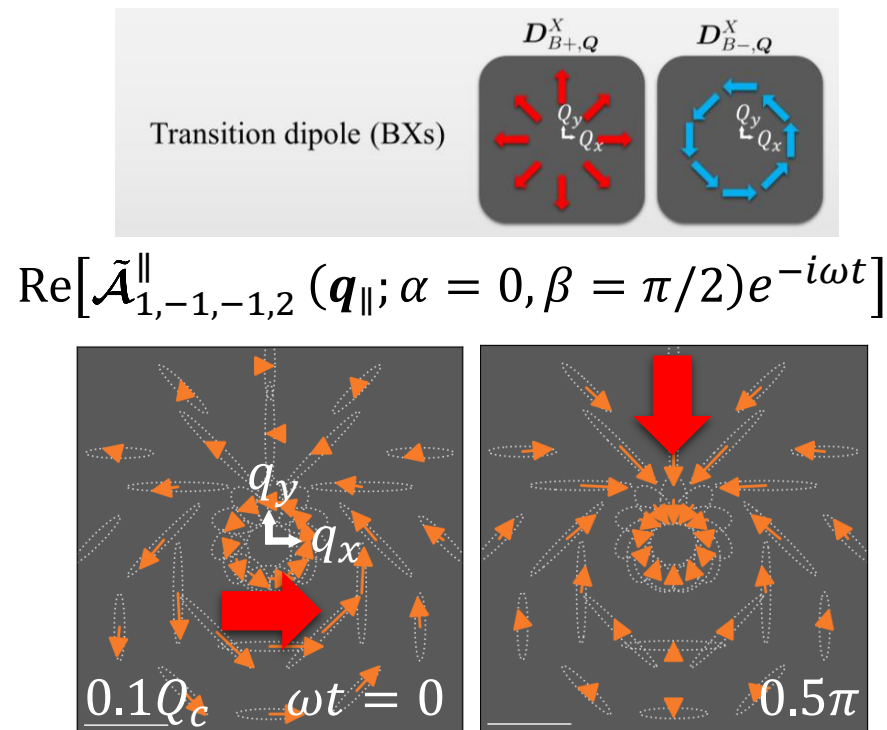


Directional selective excitation

Relax the condition $\tilde{F}_{|\ell|}(q_{\parallel}) \approx \tilde{F}_{|\ell'|}(q_{\parallel})$

Directional selective excitation
can happen when $\sigma, \ell, -\sigma, \ell' = 1, -1, -1, 2$

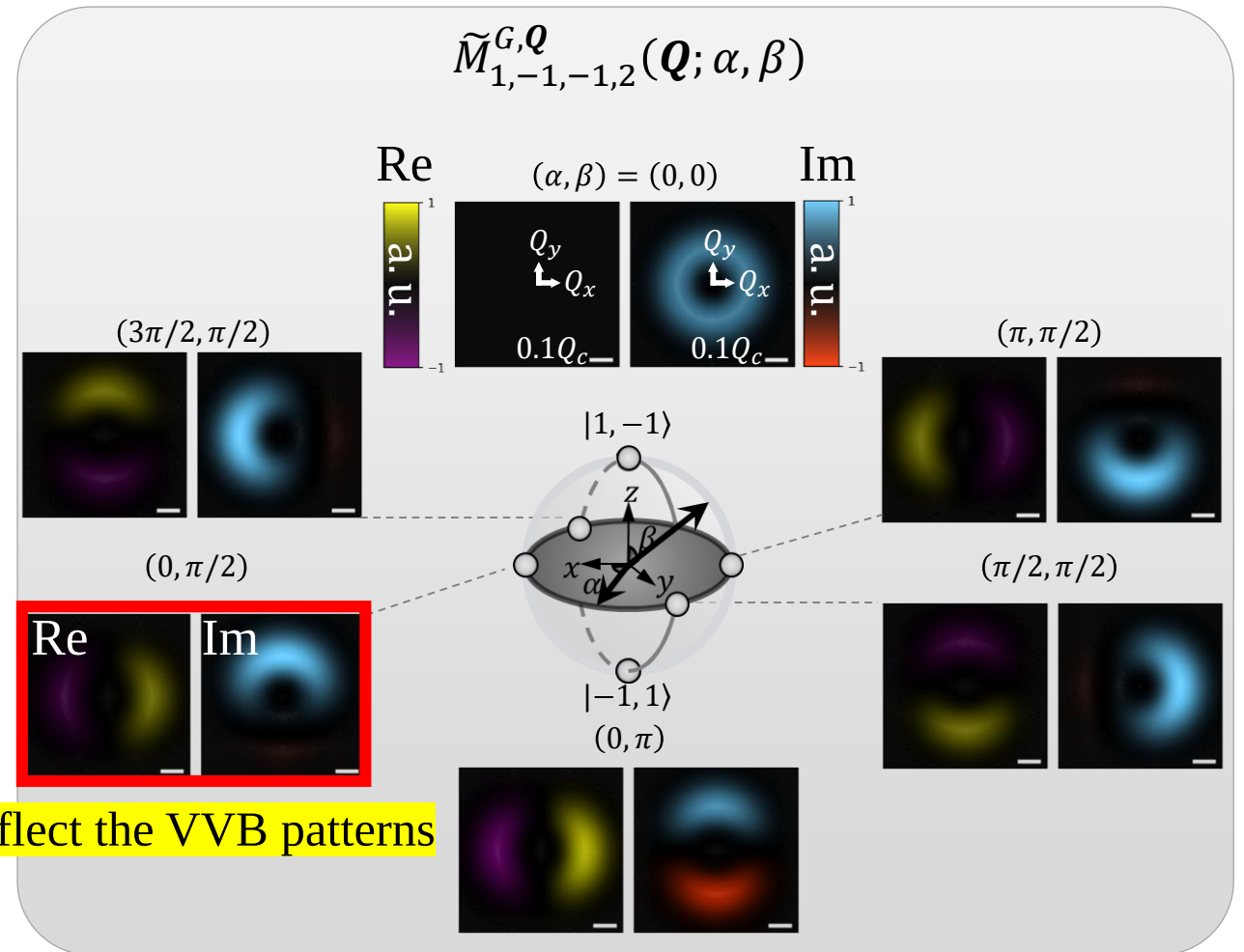
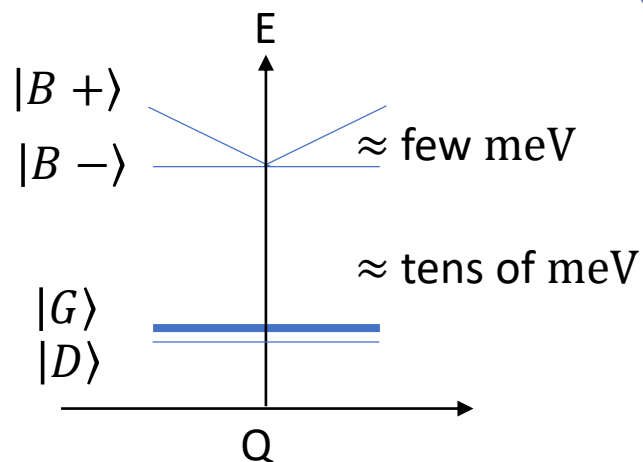
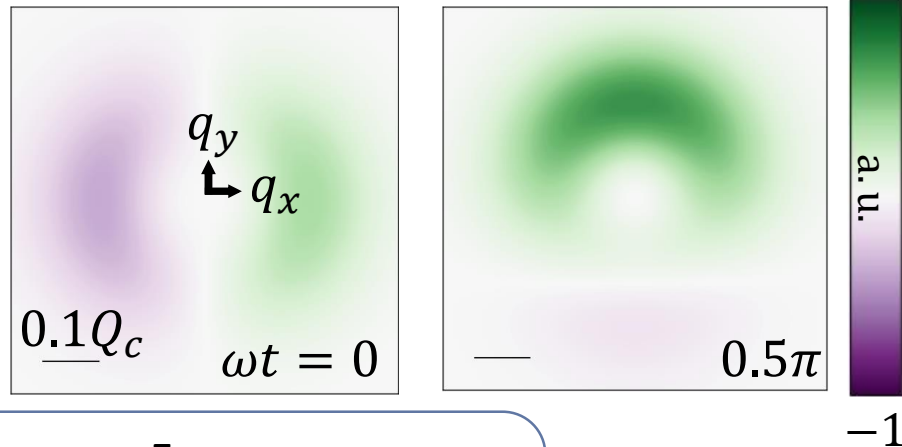
Directional selective excitation



Directional selective excitation

Transition dipole (GX) $\mathbf{D}^{G,Q} = D^G \hat{\mathbf{z}}$

$$\text{Re}[\tilde{\mathcal{A}}_{1,-1,-1,2}^Z(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \pi/2) e^{-i\omega t}]$$



1. Introduction
2. Spin-orbital angular momentum interaction in light
3. Light-matter interaction
4. Exciton wave packet
 - Global phase problem
 - Zeroth-order approximation

Summary and outlook

4. Global phase problem

Following G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019),
the superposition of finite-momentum exciton state

$$|\Psi(t)\rangle = |GS\rangle + \sum_{S=B\pm} \sum_{\mathbf{Q}} \tilde{c}^{S,\mathbf{Q}(1)}(t) e^{-i\omega^{S,\mathbf{Q}}t} |\Psi^{S,\mathbf{Q}}\rangle$$

$$\tilde{c}^{S,\mathbf{Q}(1)}(t) \approx \tilde{\gamma}^0(t) \left(\frac{\tilde{M}_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}}}{\hbar\omega^0} \right)$$

Exciton wave packet

$$f_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}}(\mathbf{R}_c, t) \equiv \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{Q}} \tilde{c}^{S,\mathbf{Q}(1)}(t) e^{i\mathbf{Q}\cdot\mathbf{R}_c}$$

$$f_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}}(\mathbf{R}_c) = \frac{1}{\hbar\omega^0\sqrt{\Omega}} \sum_{\mathbf{Q}} \tilde{M}_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}} e^{i\mathbf{Q}\cdot\mathbf{R}_c}$$

Global phase from valley-mixed BXs

$$|\Psi^{B\pm,\mathbf{Q}}\rangle = e^{i\delta(\mathbf{Q})} (e^{-i\phi_{\mathbf{Q}}} |\Psi^{K,\mathbf{Q}}\rangle \pm e^{i\phi_{\mathbf{Q}}} |\Psi^{K',\mathbf{Q}}\rangle) / \sqrt{2}$$

$$\tilde{M}_{\hat{\varepsilon},\ell}^{B\pm,\mathbf{Q}} = e^{i\delta(\mathbf{Q})} (e^{-i\phi_{\mathbf{Q}}} \tilde{M}_{\hat{\varepsilon},\ell}^{K,\mathbf{Q}} \pm e^{i\phi_{\mathbf{Q}}} \tilde{M}_{\hat{\varepsilon},\ell}^{K',\mathbf{Q}}) / \sqrt{2}$$

New strategy

If we can find this relation for exciton :

$$\Psi^{S,\mathbf{Q}}(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q}\cdot\mathbf{R}_c} U^{S,\mathbf{0}}(\mathbf{R}_c, \mathbf{r})$$

Exciton wave packet

$$\Phi_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}}(\mathbf{R}_c, \mathbf{r}, t) \equiv \sum_{\mathbf{Q}} \tilde{c}^{S,\mathbf{Q}(1)}(t) \Psi^{S,\mathbf{Q}}(\mathbf{R}_c, \mathbf{r})$$

$$\Phi_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}}(\mathbf{R}_c, \mathbf{r}, t) = \frac{U^{S,\mathbf{0}}(\mathbf{R}_c, \mathbf{r})}{\hbar\omega^0} \sum_{\mathbf{Q}} \tilde{M}_{\hat{\varepsilon},\ell}^{S,\mathbf{Q}} e^{i\mathbf{Q}\cdot\mathbf{R}_c}$$

4. Wave functions of finite-momentum excitons

Goal : $\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} U^{S,\mathbf{Q}}(\mathbf{R}_c, \mathbf{r})$

Write down the exciton wave function

$$\Psi^{S,Q}(\mathbf{r}_e, \mathbf{r}_h) = \sum_{vck} \Lambda^{S,Q}(vck) \psi_{c,k+Q}(\mathbf{r}_e) \psi_{c,k}^*(\mathbf{r}_h)$$

$$\begin{array}{l} \downarrow (\mathbf{R}_c = a\mathbf{r}_e + b\mathbf{r}_h, \mathbf{r} = \mathbf{r}_e - \mathbf{r}_h, a + b = 1) \\ \Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) = \sum_{vck} \Lambda^{S,Q}(vck) \psi_{c,k+Q}(\mathbf{R}_c + b\mathbf{r}) \psi_{c,k}^*(\mathbf{R}_c - a\mathbf{r}) \end{array}$$

$$\begin{array}{l} \downarrow (\text{Bloch theory : } \psi_{n,k}(\mathbf{r}') = u_{n,k}(\mathbf{r}') e^{i\mathbf{k} \cdot \mathbf{r}'}) \\ \boxed{\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) = e^{i\mathbf{Q} \cdot \mathbf{R}_c} U^{S,\mathbf{Q}}(\mathbf{R}_c, \mathbf{r})} \text{ Similar form} \end{array}$$

where $U^{S,Q}(\mathbf{R}_c, \mathbf{r}) = \sum_{vck} \Lambda^{S,Q}(vc \mathbf{k} - b\mathbf{Q}) e^{i\mathbf{k} \cdot \mathbf{r}} u_{c,k+aQ}(\mathbf{R}_c + b\mathbf{r}) u_{c,k-bQ}^*(\mathbf{R}_c - a\mathbf{r})$

$$\begin{array}{l} \downarrow (\text{choose } b = 0, a = 1) \\ \equiv \sum_{vck} e^{i\mathbf{k} \cdot \mathbf{r}} \Lambda^{S,\mathbf{Q}}(vck) u_{c,k+\mathbf{Q}}(\mathbf{R}_c) u_{c,k}^*(\mathbf{R}_c - \mathbf{r}) \end{array}$$

4. Zeroth-order approximation

Goal : $\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} U^{S,0}(\mathbf{R}_c, \mathbf{r})$

We have : $\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) = e^{i\mathbf{Q} \cdot \mathbf{R}_c} U^{S,Q}(\mathbf{R}_c, \mathbf{r})$, where $U^{S,Q}(\mathbf{R}_c, \mathbf{r}) = \sum_{vck} e^{i\mathbf{k} \cdot \mathbf{r}} \boxed{\Lambda^{S,Q}(vck) u_{c,k+Q}(\mathbf{R}_c)} u_{c,k}^*(\mathbf{R}_c - \mathbf{r})$

Using **Taylor expansion** around $\mathbf{Q} = 0$

$$\Lambda^{S,Q}(vck) u_{c,k+Q}(\mathbf{R}_c) = \underbrace{\Lambda^{S,0}(vck) u_{c,k}(\mathbf{R}_c)}$$

Zeroth-order

$$+ \underbrace{\Lambda^{S,0}(vck) [\mathbf{Q} \cdot \nabla_k u_{c,k}(\mathbf{R}_c)] + [\mathbf{Q} \cdot \nabla_k \Lambda^{S,0}(vck)] u_{c,k}(\mathbf{R}_c)}$$

First-order

$$+ \underbrace{[\mathbf{Q} \cdot \nabla_k \Lambda^{S,0}(vck)] [\mathbf{Q} \cdot \nabla_k u_{c,k}(\mathbf{R}_c)] + \dots}_{\text{Second-order}} + \text{higher-order terms}$$

Second-order

4. Valley Excitons

$$\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} U^{S,0}(\mathbf{R}_c, \mathbf{r})$$

Exciton wave packet $\Phi_{\hat{\mathbf{e}},\ell}^{K/K',Q}(\mathbf{R}_c, \mathbf{r}) = \frac{U^{K/K',0}(\mathbf{R}_c, \mathbf{r})}{\hbar\omega^0} \sum_Q \tilde{M}_{\hat{\mathbf{e}},\ell}^{K/K',Q} e^{i\mathbf{Q} \cdot \mathbf{R}_c}$

$$U^{S,0}(\mathbf{R}_c, \mathbf{r}) \approx \sum_{vck} e^{ik \cdot \mathbf{r}} \Lambda^{S,0}(vck) u_{c,k}(\mathbf{R}_c) u_{c,k}^*(\mathbf{R}_c - \mathbf{r})$$

1. Recombination

$$\mathbf{r} = \mathbf{0}$$

2. 2D hydrogen model

$$\Lambda^{1s,0}(vck) = \frac{\sqrt{2\pi} a_B^*}{\left[1 + \left(\frac{a_B^* k}{2}\right)^2\right]^{3/2}}$$

Real function ($a_B^* \approx 1\text{nm}$)

3. Only show the wave packet on lattice site.

$$u_{n,k}(\mathbf{r}') = u_{n,k}(\mathbf{r}' + \mathbf{R})$$

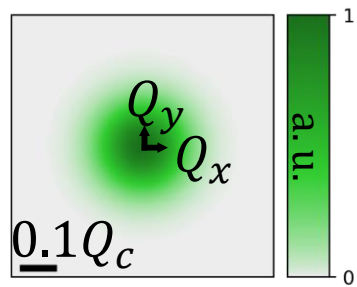
$\mathbf{R}$ (Bravais lattice vector) $\ll W_0$ (Bema waist)

4. Valley Excitons

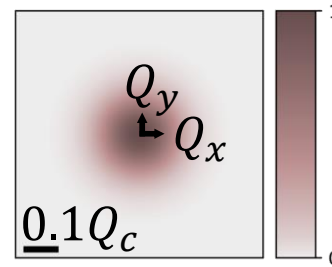
Exciton wave packet $\Phi_{\hat{\mathbf{e}},\ell}^{K/K',\mathbf{Q}}(\mathbf{R}_c, \mathbf{r}) = \frac{U^{K/K',0}(\mathbf{R}_c, \mathbf{r})}{\hbar\omega^0} \sum_{\mathbf{Q}} \tilde{M}_{\hat{\mathbf{e}},\ell}^{K/K',\mathbf{Q}} e^{i\mathbf{Q}\cdot\mathbf{R}_c}$

Ex. $\ell = 0$

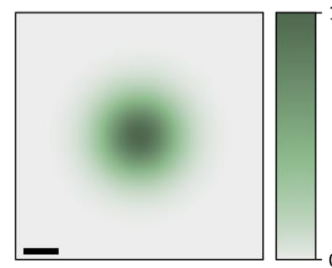
$$|\tilde{F}_{|\ell=0|}(\mathbf{Q})e^{i\ell\phi_{\mathbf{Q}}}|^2$$



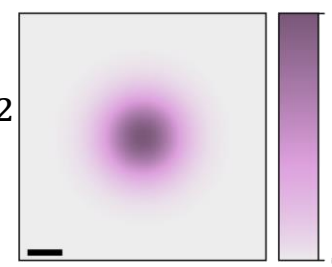
$$|\tilde{M}_{\hat{\mathbf{x}},0}^{K,\mathbf{Q}}|^2$$



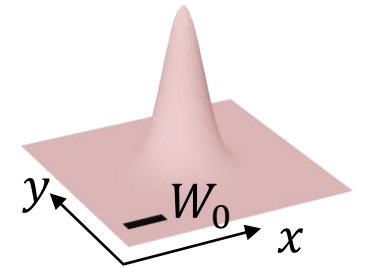
$$|\tilde{M}_{\hat{\mathbf{x}},0}^{K',\mathbf{Q}}|^2$$



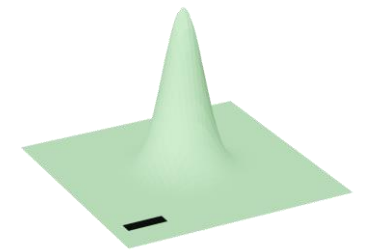
$$|\tilde{M}_{(\hat{\mathbf{x}}+i\hat{\mathbf{y}})/\sqrt{2},0}^{S,\mathbf{Q}}|^2$$



$$|\Phi_{\hat{\mathbf{x}},0}^K(\mathbf{R}_c, \mathbf{r} = \mathbf{0})|^2$$



$$|\Phi_{\hat{\mathbf{x}},0}^{K'}(\mathbf{R}_c, \mathbf{r} = \mathbf{0})|^2$$



$$|\Phi_{(\hat{\mathbf{x}}+i\hat{\mathbf{y}})/\sqrt{2},0}^{K/K'}(\mathbf{R}_c, \mathbf{r} = \mathbf{0})|^2$$

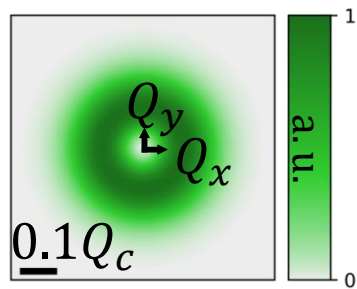


4. Valley Excitons

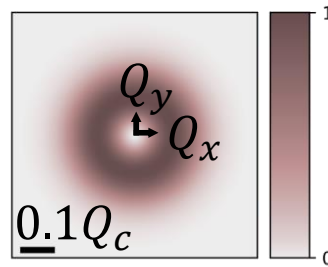
Exciton wave packet $\Phi_{\hat{\mathbf{e}},\ell}^{K/K',Q}(\mathbf{R}_c, \mathbf{r}) = \frac{U^{K/K',0}(\mathbf{R}_c, \mathbf{r})}{\hbar\omega^0} \sum_{\mathbf{Q}} \tilde{M}_{\hat{\mathbf{e}},\ell}^{K/K',Q} e^{i\mathbf{Q}\cdot\mathbf{R}_c}$

Ex. $\ell = \pm 1$

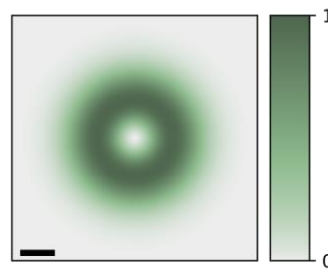
$$|\tilde{F}_{|\ell=\pm 1|}(\mathbf{Q})e^{i\ell\phi_{\mathbf{Q}}}|^2$$



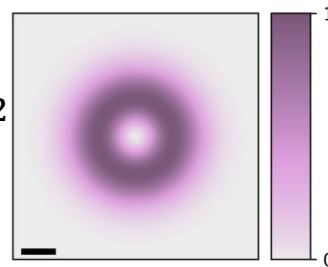
$$|\tilde{M}_{\hat{\mathbf{x}},\pm 1}^{K,Q}|^2$$



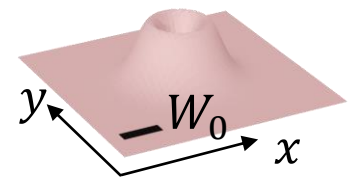
$$|\tilde{M}_{\hat{\mathbf{x}},\pm 1}^{K',Q}|^2$$



$$|\tilde{M}_{(\hat{\mathbf{x}}+i\hat{\mathbf{y}})/\sqrt{2},\pm 1}^{S,Q}|^2$$



$$|\Phi_{\hat{\mathbf{x}},\pm 1}^K(\mathbf{R}_c, \mathbf{r} = \mathbf{0})|^2$$



$$|\Phi_{\hat{\mathbf{x}},\pm 1}^{K'}(\mathbf{R}_c, \mathbf{r} = \mathbf{0})|^2$$



$$|\Phi_{(\hat{\mathbf{x}}+i\hat{\mathbf{y}})/\sqrt{2},\pm 1}^{K/K'}(\mathbf{R}_c, \mathbf{r} = \mathbf{0})|^2$$



4. Valley-mixed Excitons

Exciton wave packet $\Phi_{\hat{\varepsilon},\ell}^{B\pm,Q}(\mathbf{R}_c, \mathbf{r}) =$

$$(2\hbar\omega^0)^{-1} |U^{S,0}(\mathbf{R}_c, \mathbf{r})| \left[\boxed{e^{i\text{Arg}[U^{S,0}(\mathbf{R}_c, \mathbf{r})]}} \sum_{\mathbf{Q}} \left(e^{-i\ell\phi_{\mathbf{Q}}} \tilde{M}_{\hat{\varepsilon},\ell}^{K,\mathbf{Q}} \pm e^{i\ell\phi_{\mathbf{Q}}} \tilde{M}_{\hat{\varepsilon},\ell}^{K',\mathbf{Q}} \right) e^{i\mathbf{Q}\cdot\mathbf{R}_c} e^{-i\ell\phi_{\mathbf{Q}}} \right. \\ \left. \pm \boxed{e^{-i\text{Arg}[U^{S,0}(\mathbf{R}_c, \mathbf{r})]}} \sum_{\mathbf{Q}} \left(e^{-i\ell\phi_{\mathbf{Q}}} \tilde{M}_{\hat{\varepsilon},\ell}^{K,\mathbf{Q}} \pm e^{i\ell\phi_{\mathbf{Q}}} \tilde{M}_{\hat{\varepsilon},\ell}^{K',\mathbf{Q}} \right) e^{i\mathbf{Q}\cdot\mathbf{R}_c} e^{i\ell\phi_{\mathbf{Q}}} \right]$$

Further investigate in
phase structure

Summary and outlook

Summary :

- In the specific case of VVB, the phase and polarization structures can be independently controlled by the average and difference of the angular momenta of the constituent beams.
- VVB with unequal OAM magnitudes ($|\ell| \neq |\ell'|$) produce asymmetric transverse polarization and longitudinal component, leading to asymmetric exciton transition-rate patterns in center-of-mass momentum space.
- For valley-mixed exciton wave packets, the phase of the exciton wave function must be considered together with the optical phase to accurately describe the wave-packet structure.

Outlook

- Selective excitation of excitons: Use tailored vector vortex beams to selectively drive valley-mixed bright and gray excitons within specific center-of-mass momentum ranges.
- Topology of light–exciton coupling: Investigate how the topological structures of vector vortex beams are transferred to exciton wave packets and explore their implications for exciton topology.